



基礎腦科學概論

Animal models for brain science

2026.09.29

Chia-Hsiang Chang (張家祥)



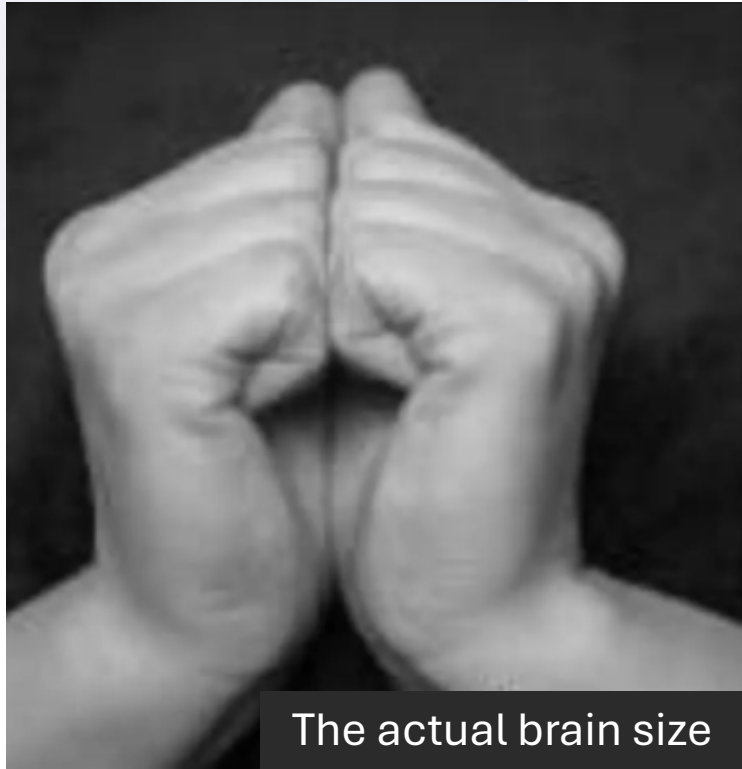
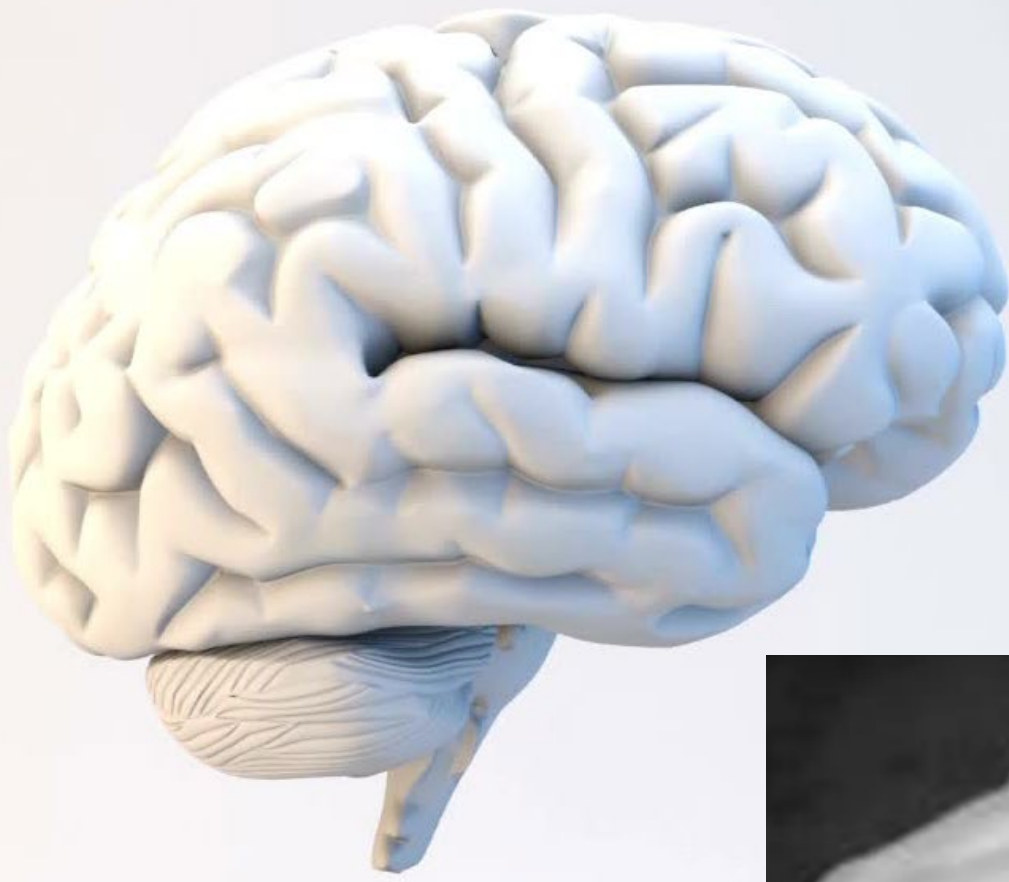
Learning Objects

- Why do we need animal models in neuroscience?
- Major animal species used in brain research.
- Introduce basic experimental techniques employed in animal models for neuroscience studies



Learning Objects

- Why do we need animal models in neuroscience?
- Major animal species used in brain research.
- Introduce basic experimental techniques employed in animal models for neuroscience studies



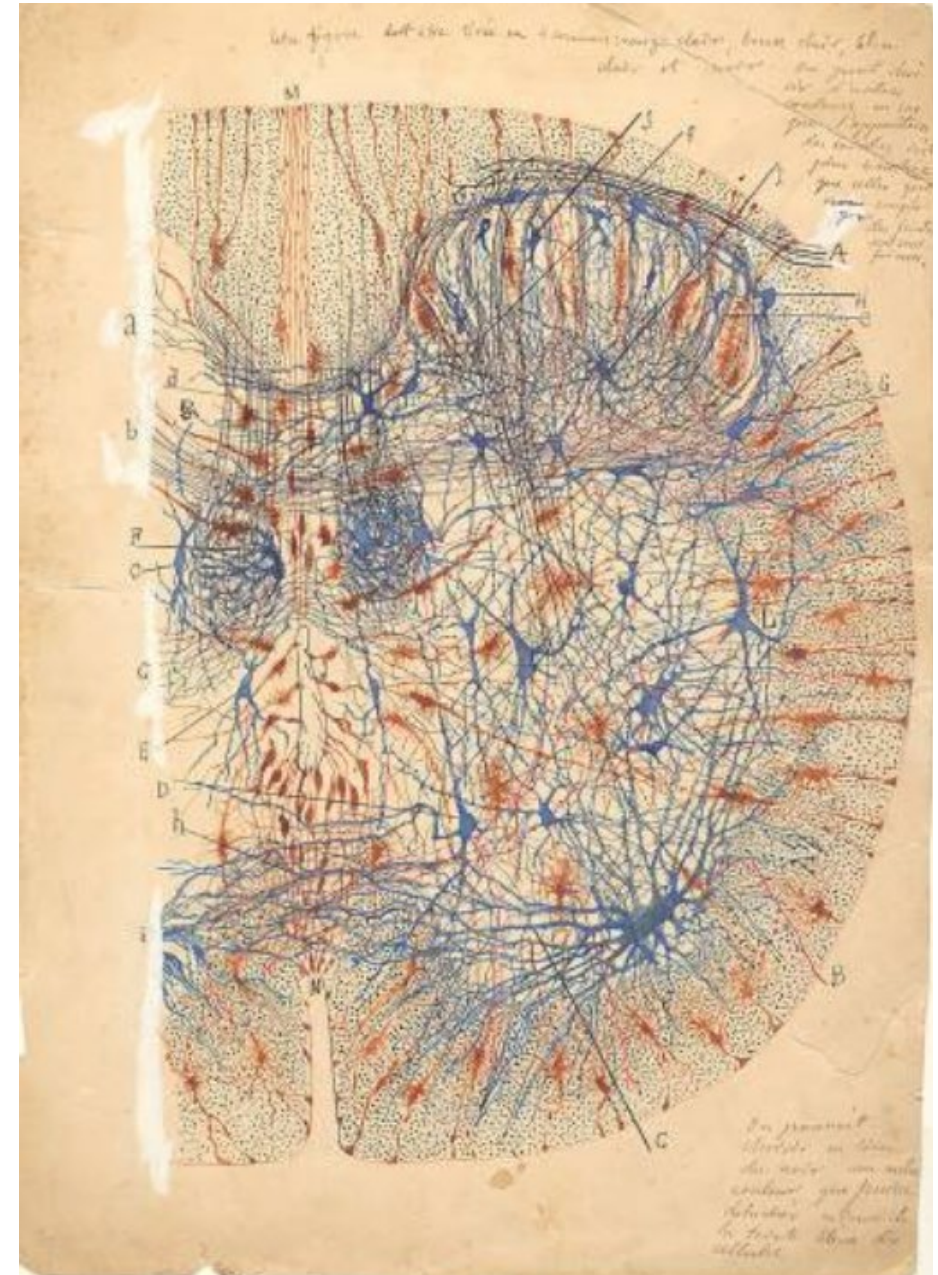
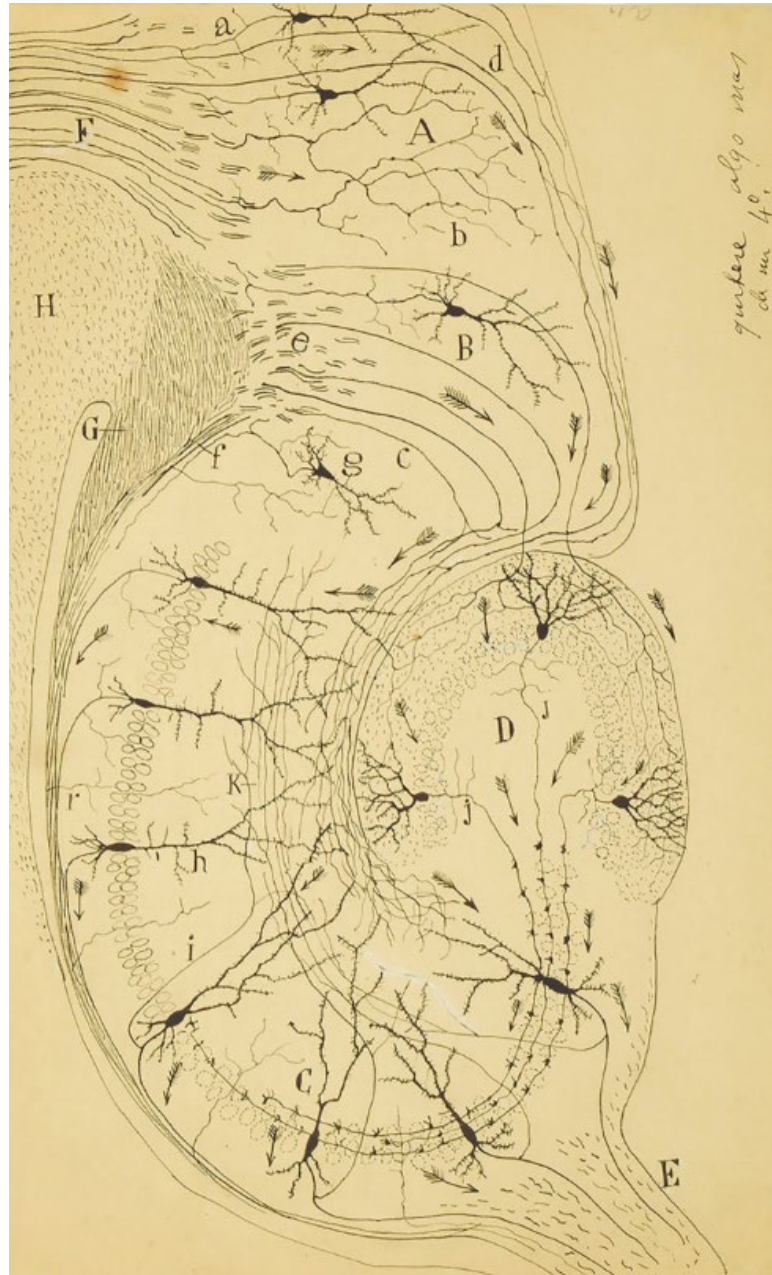
The actual brain size





Santiago Ramón y Cajal
(1852-1934)

Golgi staining

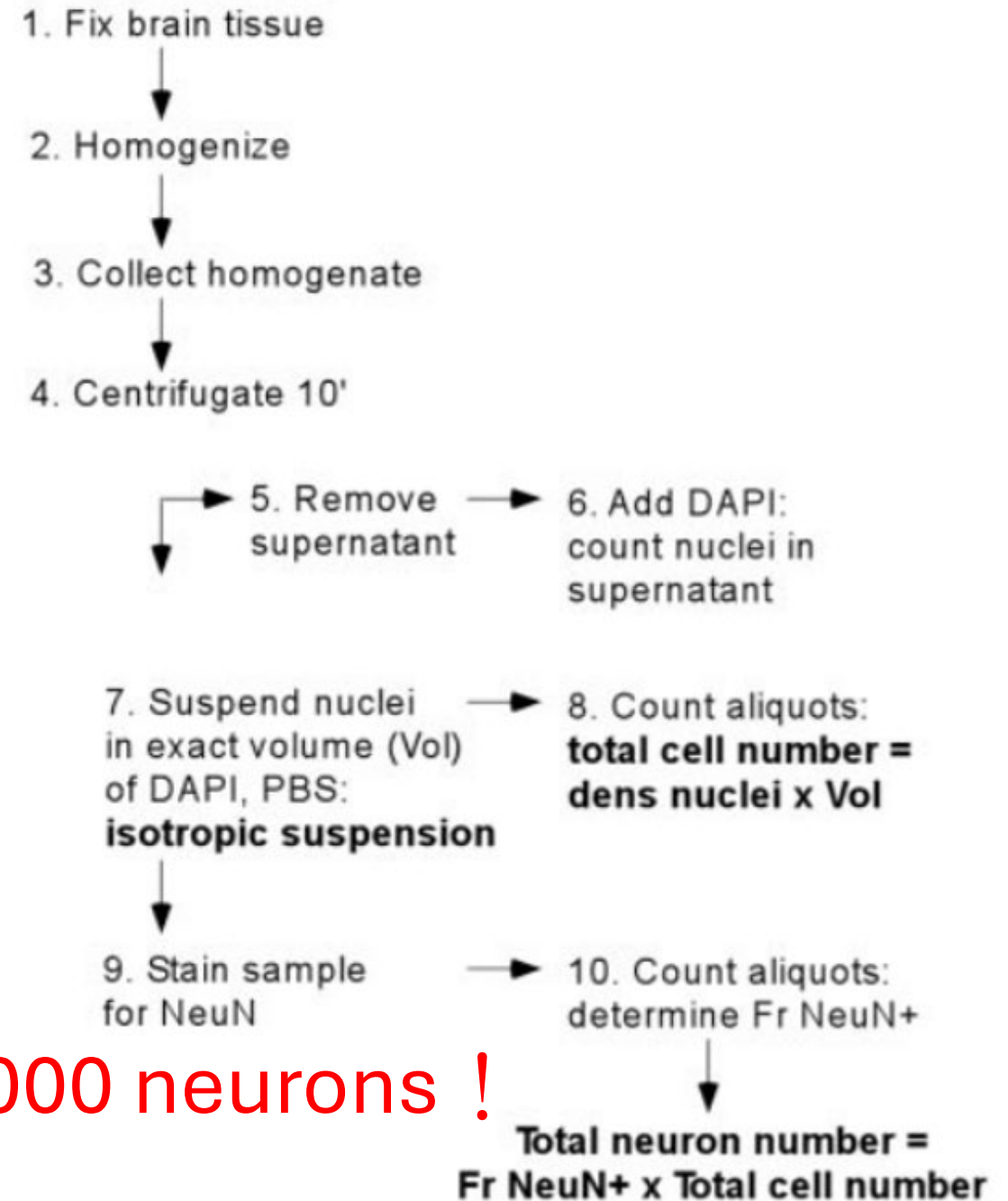
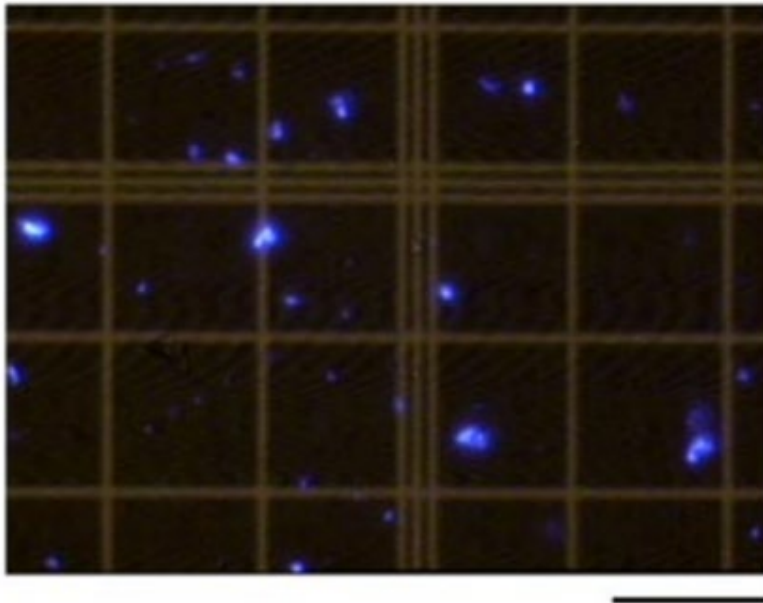


Brief Communication

Isotropic Fractionator: A Simple, Rapid Method for the Quantification of Total Cell and Neuron Numbers in the Brain

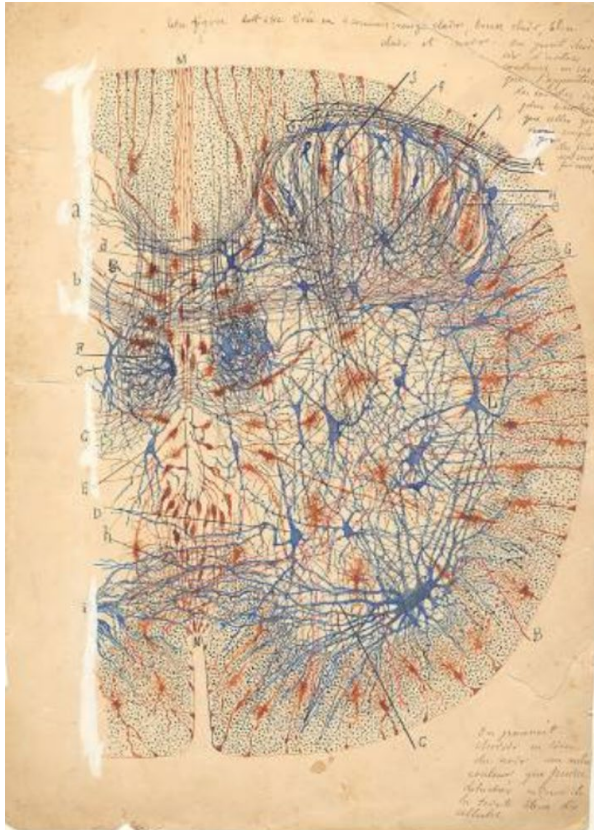
Suzana Herculano-Houzel and Roberto Lent

Departamento de Anatomia, Instituto de Ciências Biomédicas, Universidade Federal do Rio de Janeiro, Rio de Janeiro, Brazil 21941-590

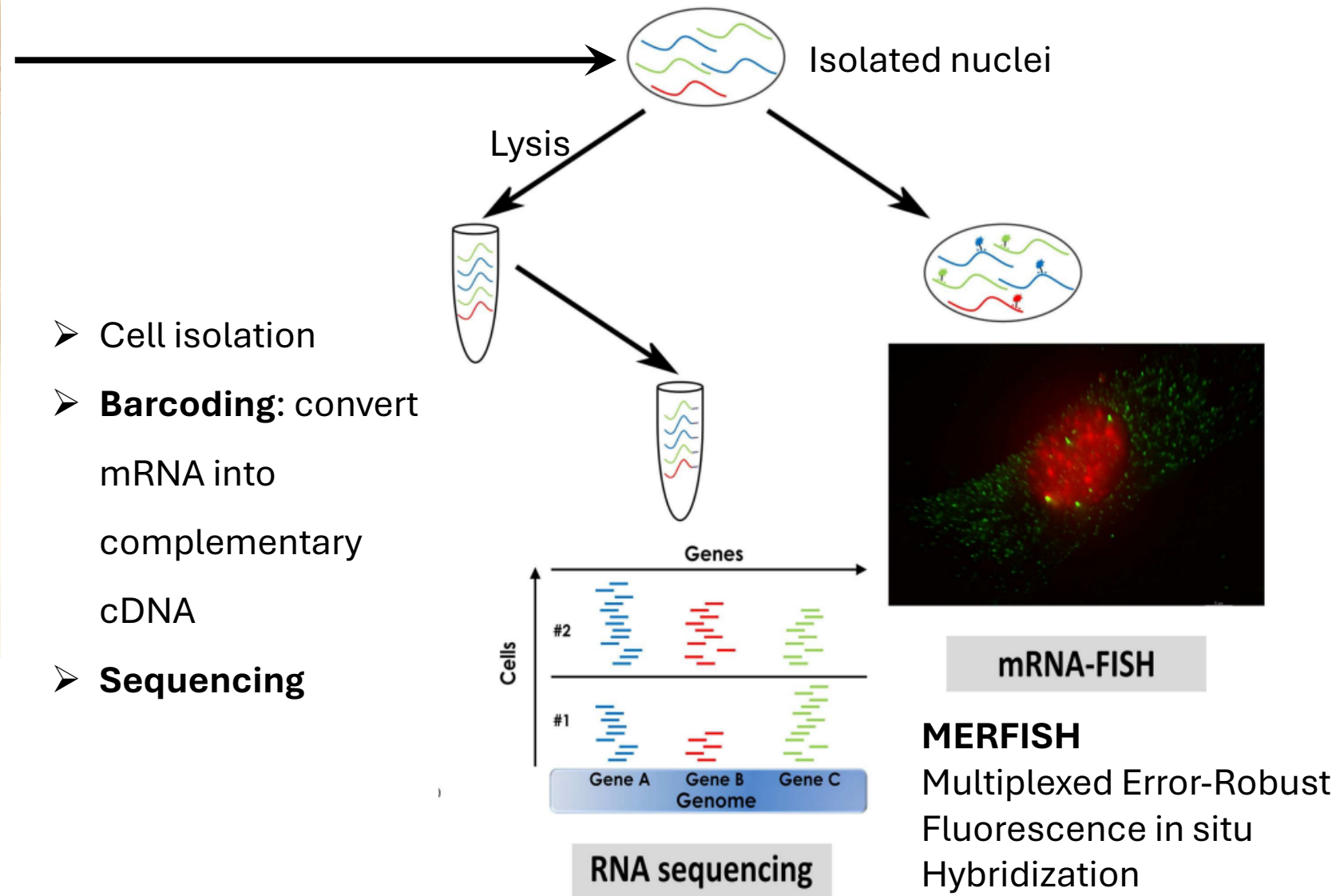


86,000,000,000 neurons !

Single-cell transcriptomics improves our understanding of cell signatures in the mammalian brain



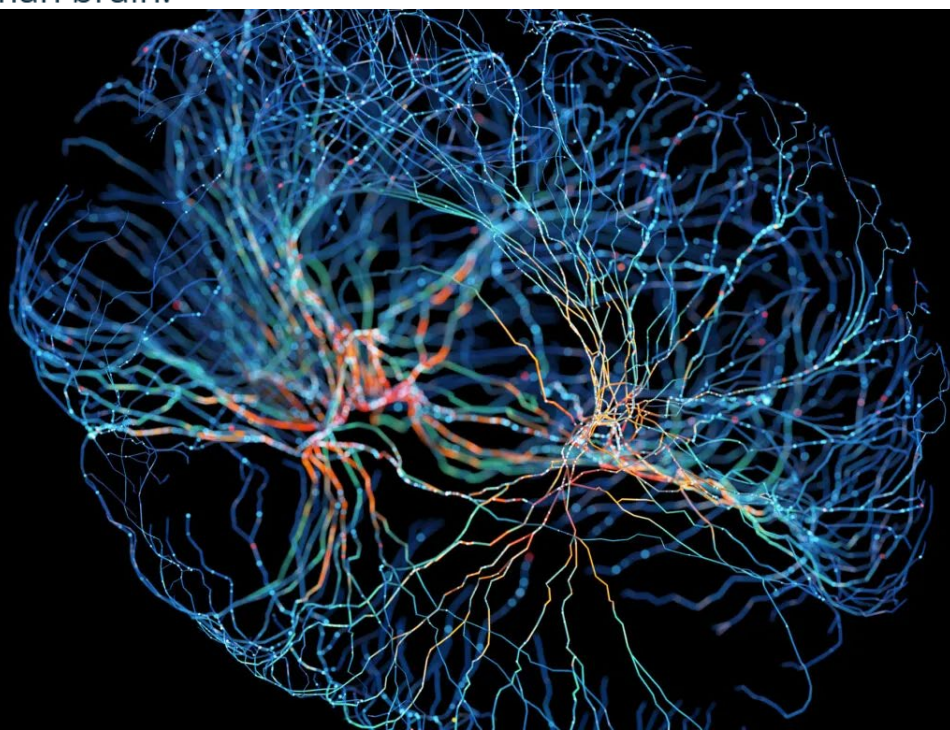
- Cell isolation
- **Barcoding**: convert mRNA into complementary cDNA
- **Sequencing**



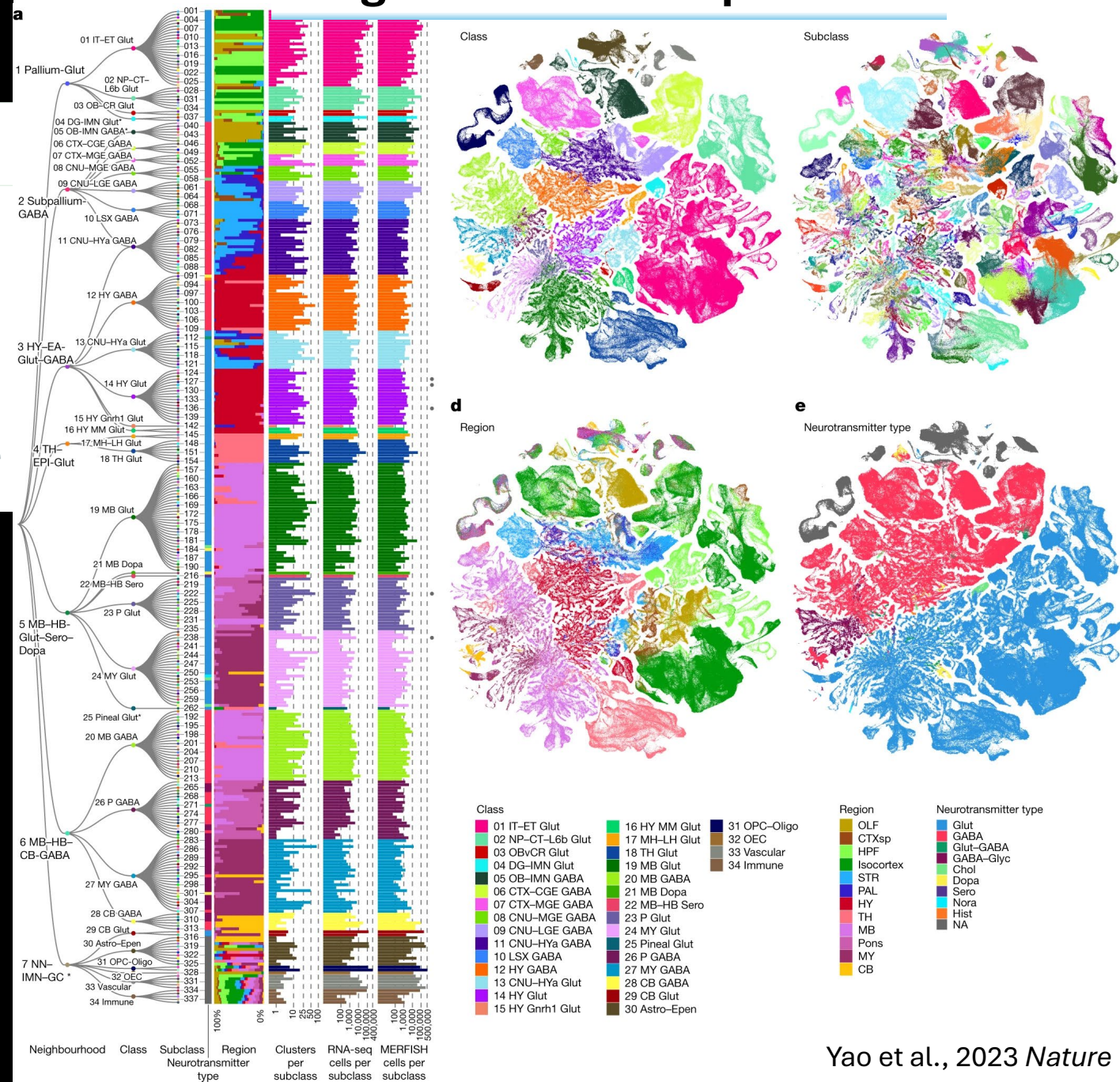
News Future tech Nature Space Health ▾ Everyday science

'A pivotal moment in neuroscience': Scientists finally discover the brain cells that make you unique

A new watershed study has mapped over 3,000 cell types in the human brain.

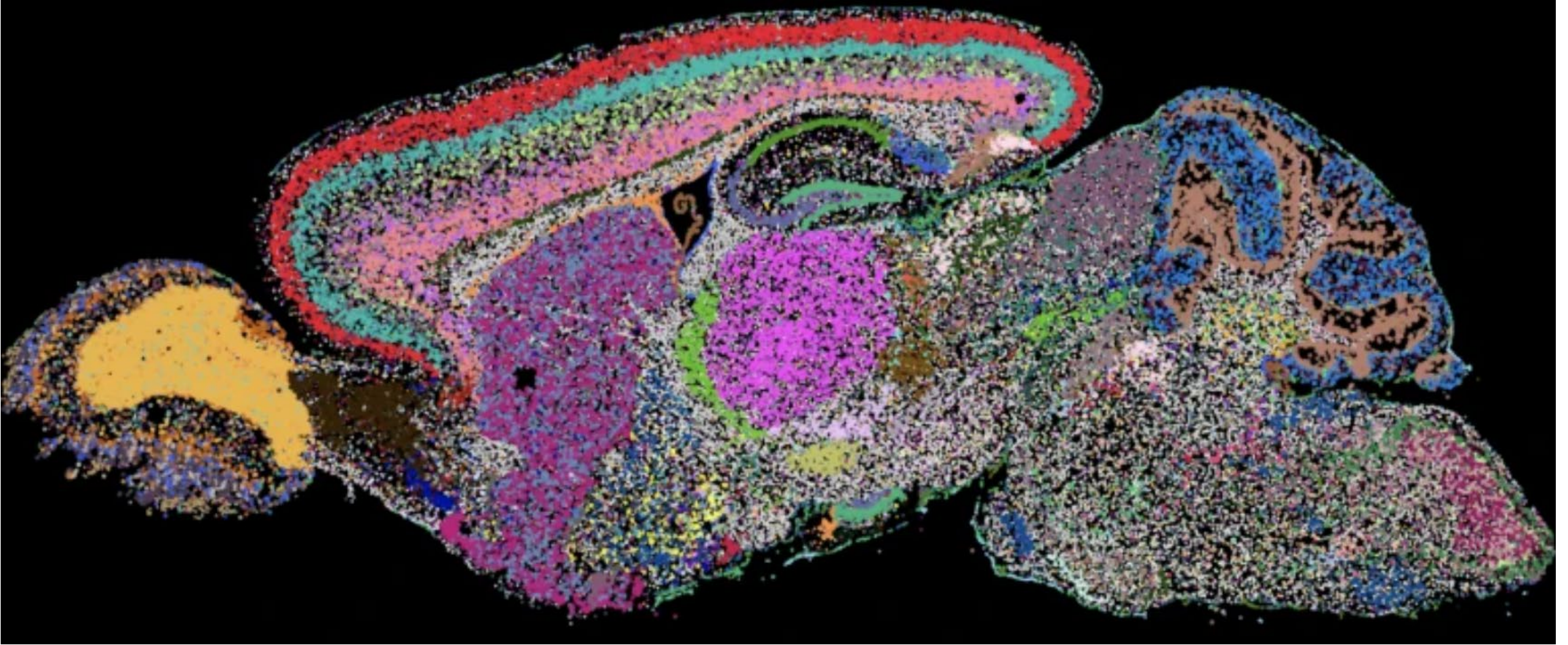


Single-cell transcriptomics



Ernest Solvay Prize honours spatial genomics pioneer

MERFISH: Multiplexed Error-Robust
Fluorescence in situ Hybridization



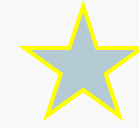
A cell atlas of a whole mouse brain generated by MERFISH. Different cell types are shown in different colours.

Credit: Xiaowei Zhuang Laboratory. Howard Hughes Medical Institute, Harvard University.

<https://www.nature.com/articles/d42473-026-00007-6>



Learning Objects



- Why do we need animal models in neuroscience?
 - ✓ Limited access to living brain tissue
 - ✓ Ethical constraints
 - ✓ Complexity of neural circuits
 - ✓ Need for controlled experimentation



Learning Objects

- Why do we need animal models in neuroscience?
- Major animal species used in brain research.
- Introduce basic experimental techniques employed in animal models for neuroscience studies



Learning Objects

- Why do we need animal models in neuroscience?
- Major animal species used in brain research.

Marine animals

Squid

Aplysia

Freshwater organisms

C. elegans

Zebrafish

Axolotl

Xenopus

Land animals

Drosophila

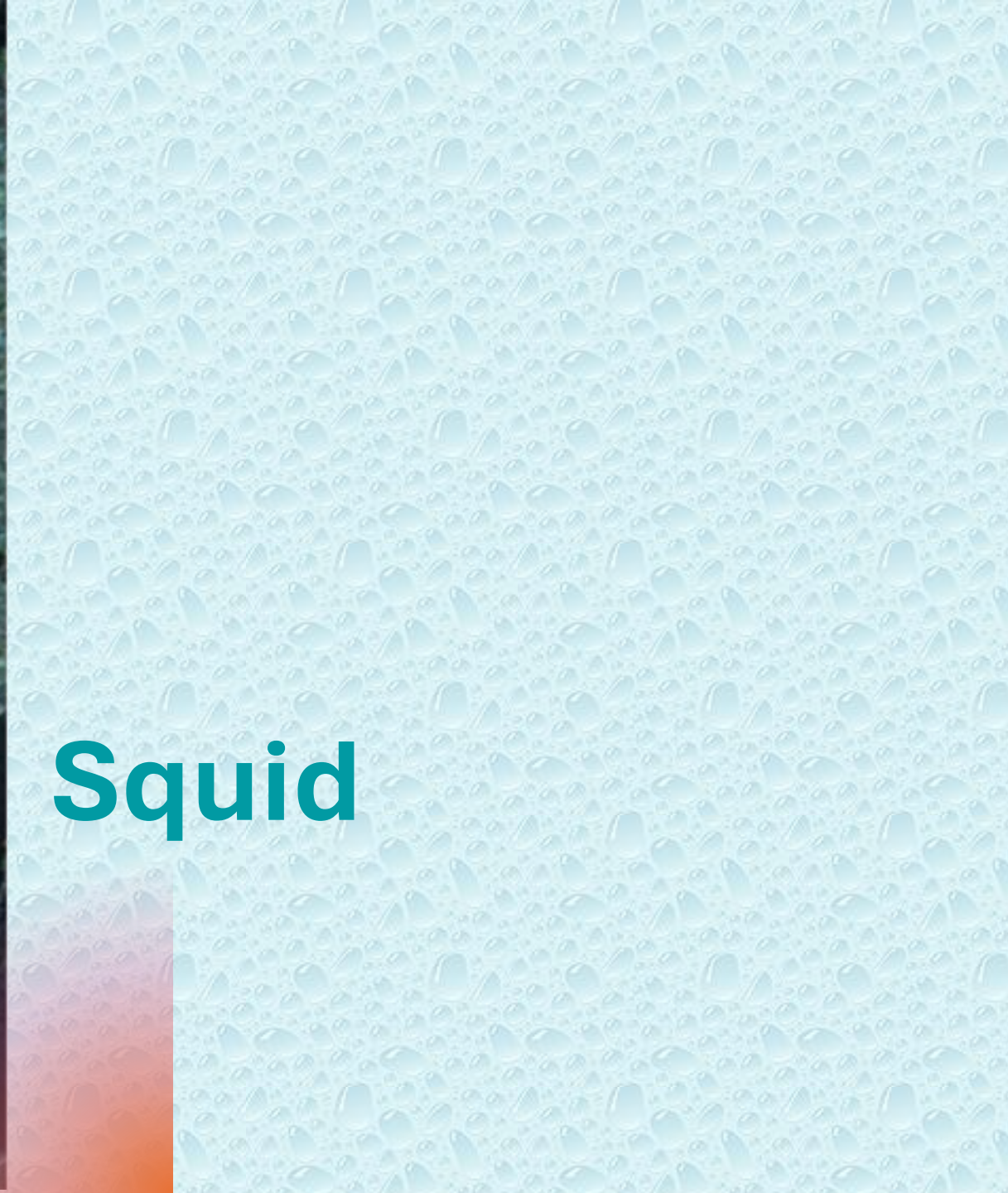
Mouse

Marmoset

Macaque



Model organism: **Squid**



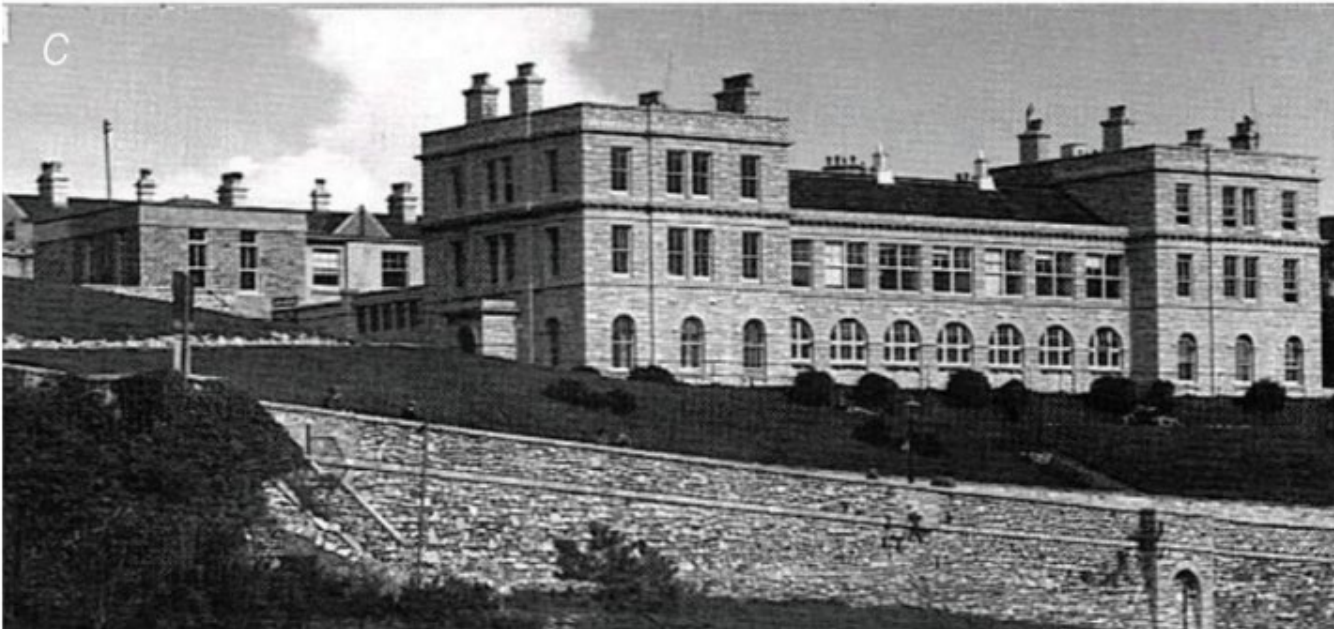


Marine Biological Association (MBA)

(A) John Zachary Young (1907–1997). The man who discovered the squid **giant axon** in the 1930s.

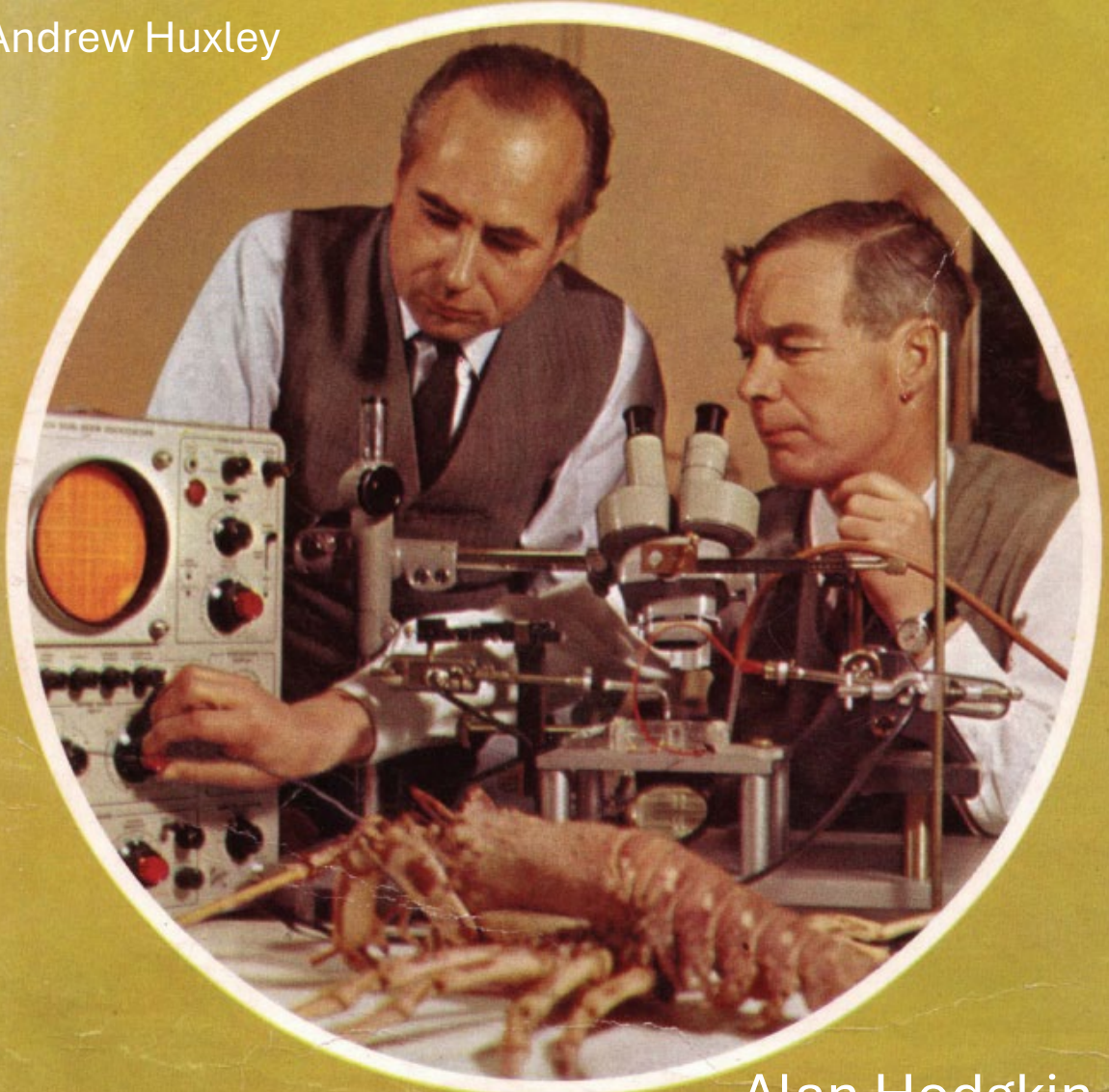
(B) *Loligo forbesi*, the long-fin squid (~60 cm long). Squid image taken by David Nicholson, courtesy of the MBA, Plymouth.

(C) The MBA on Citadel Hill in Plymouth: **first recordings** from squid giant axons.



The cover of the 1963 Nobel Prize Program

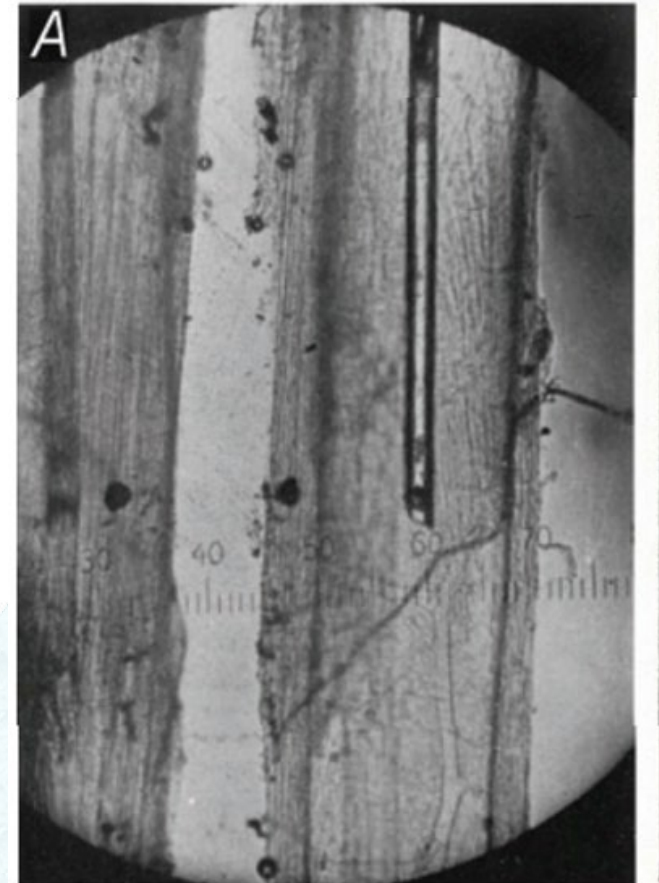
Andrew Huxley



Alan Hodgkin

NERVE-CELL ENIGMA SOLVED

The British scientists, A. L. Hodgkin and A. F. Huxley, experimenting with the nerve fibers of squids and lobsters.



Christof J. Schwiening (2012) *Journal of Physiology*

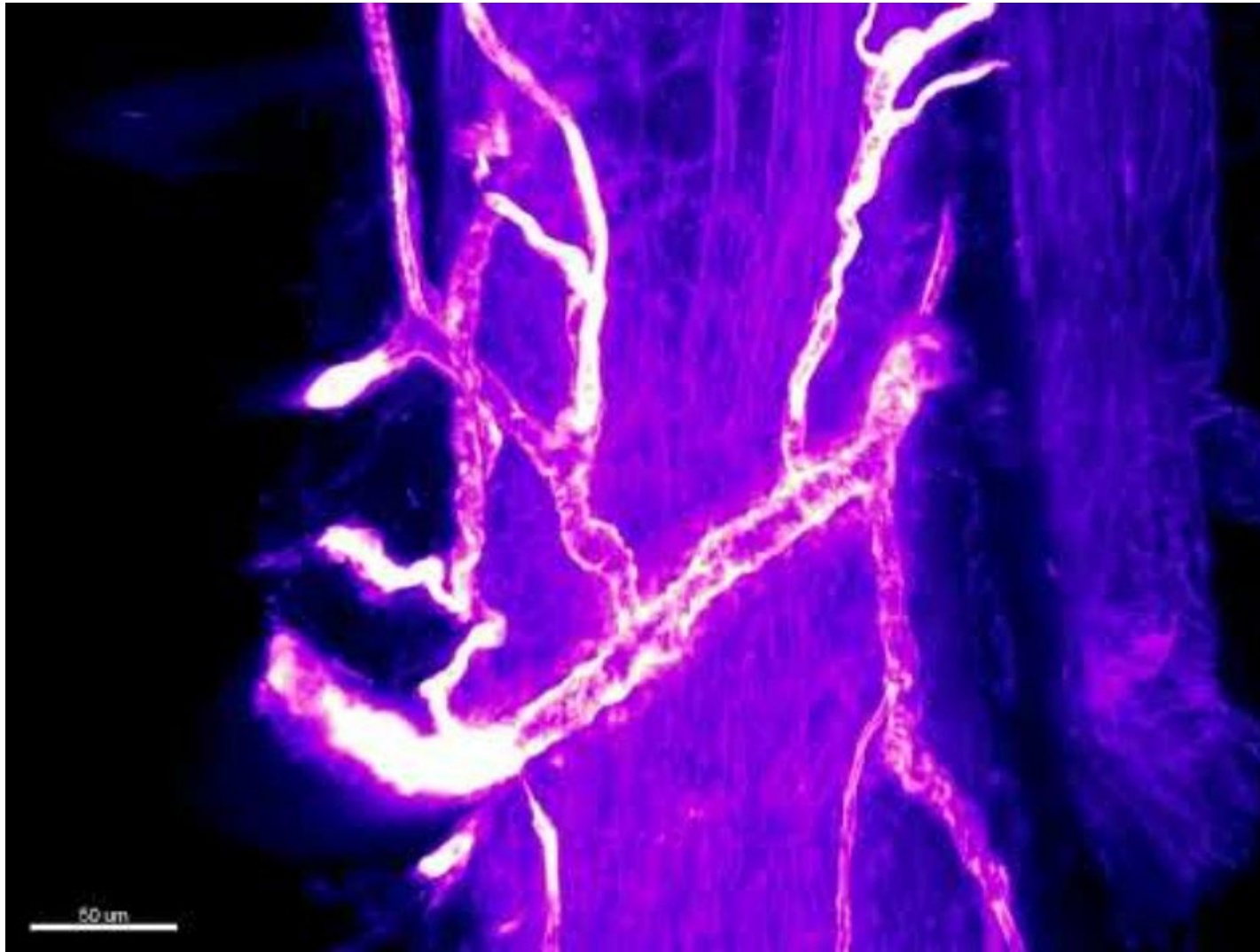


Scale Comparison

Metric	<i>Loligo forbesi</i> Giant Axon	Regular Human Axon
Typical Diameter	400-1,000 μm	0.5-2 μm
Visibility	Visible by naked eyes	Requires microscope
Function	Immediate jet-propulsion escape response	Complex sensory, motor, and cognitive functions

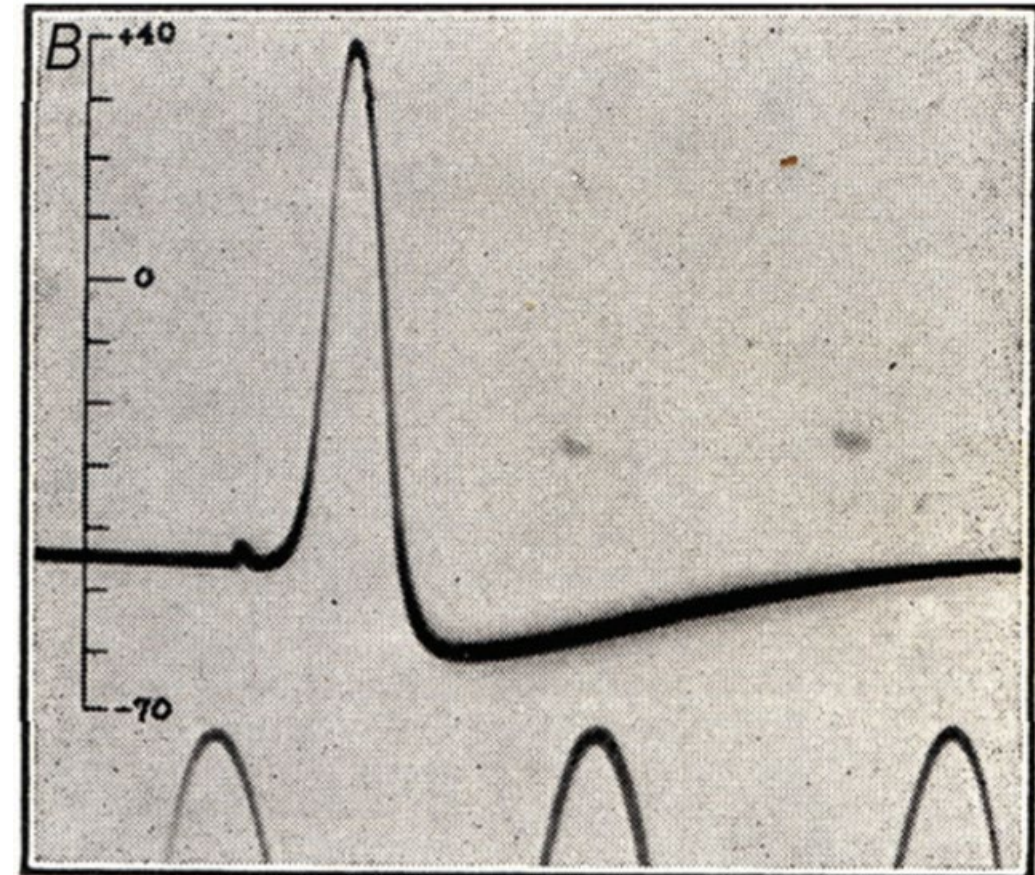
The Squid Giant Axon: A perfect model to study biological properties in **axons**

The Squid Giant Axon: A perfect model to study biological properties in axons



Squid axon and surrounding vasculatures

<https://www.youtube.com/watch?v=WBOKG0yj2DQ>



Hodgkin & Huxley, 1939, *Nature*

1939: first recordings of action potentials
1946-1952: electrophysiology

The Squid Giant Axon: A perfect model to study biological properties in axons



Ronald Vale (UCSF):
UC Santa Barbara
Marine Biological Laboratory



Organelle transport in axons using squid giant axons

Microtubule proteins, kinesins, and bi-directional movements

<https://www.youtube.com/watch?v=lwcCVp4ICWY>



Model organism:

**Sea slug
(Sea hare)**

Nobel Prize in Physiology or Medicine 2000

Learning about the Human Brain from **Sea Slugs**
(*Aplysia californica*) **California Sea hare**



Photo from the Nobel Foundation archive.

Eric R. Kandel



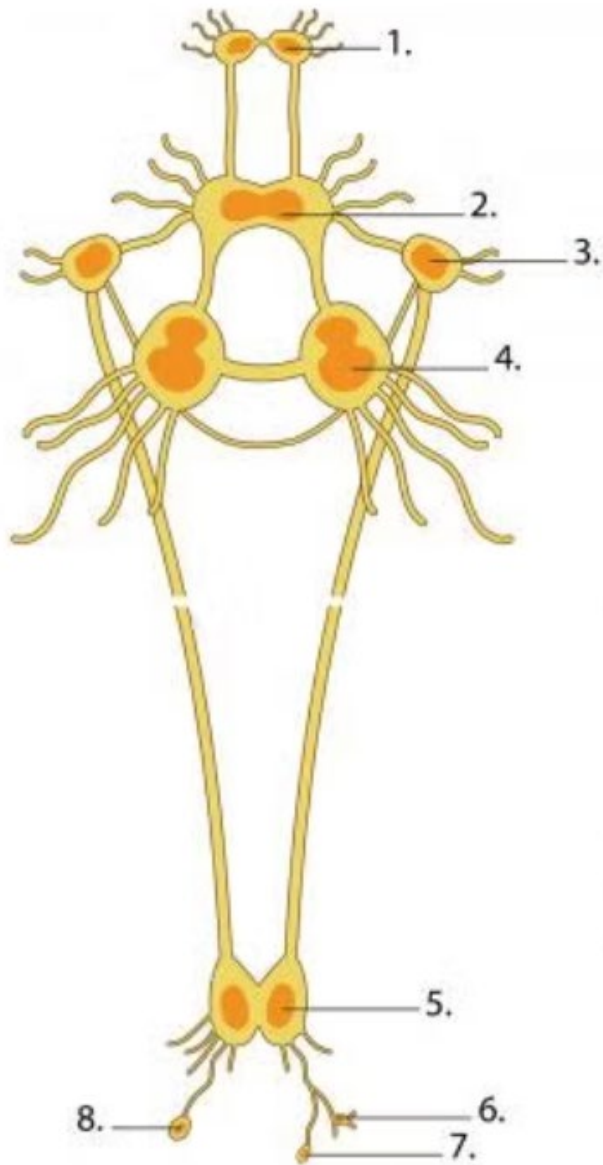
https://en.wikipedia.org/wiki/California_sea_hare

Why Sea Slugs? ★

Simple Nervous System: The sea slugs possess ~20,000 neurons

Large Neurons: Their individual neurons are large (Diameter: 100 to 1,000 μm , compared to 10-20 μm in human neurons)

Identifiable Circuits: Specific cells and their connections are mostly identical across different sea slugs.



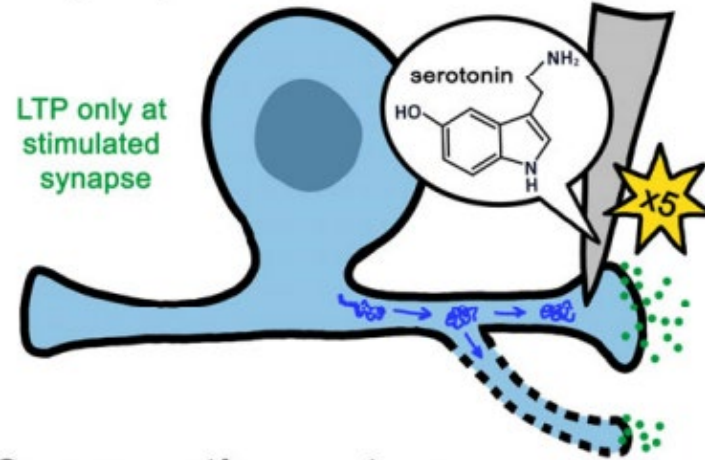
1. Buccal ganglion.
2. Cerebral ganglion.
3. Pleural ganglion.
4. Pedal ganglion.
5. Abdominal ganglion.
6. Branchial ganglion.
7. Osphradial ganglion.
8. Genital ganglion.

Credit: Anastasiya Kurylyuk from Chiel Lab.

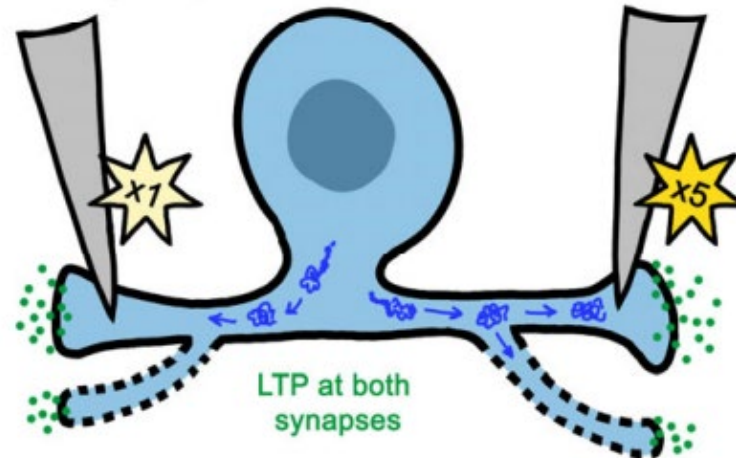
The first evidence of learning and memory: Using sea slugs as animal models



1. synapse-specific LTP



2. synaptic capture



STIMULATION

cAMP

PKA

MAPK

CREB-1

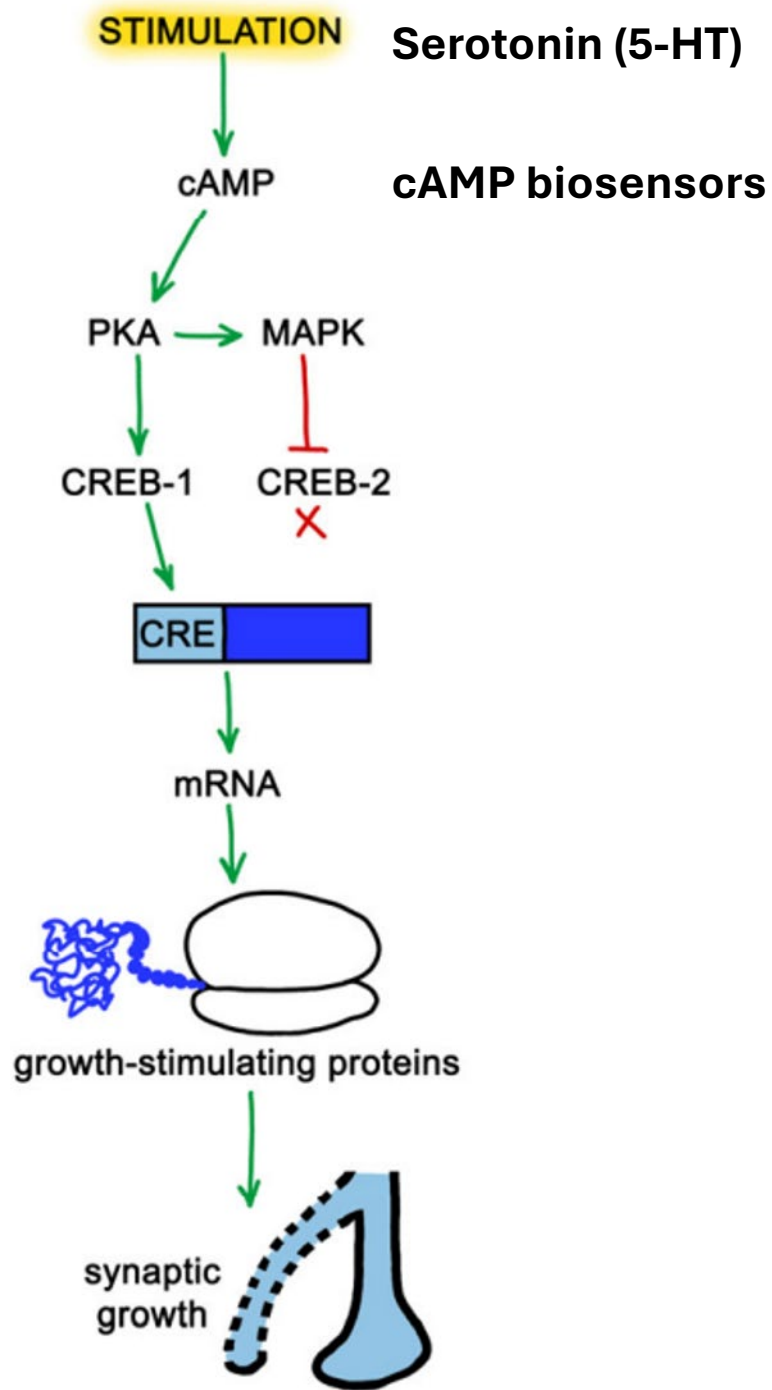
CREB-2

CRE

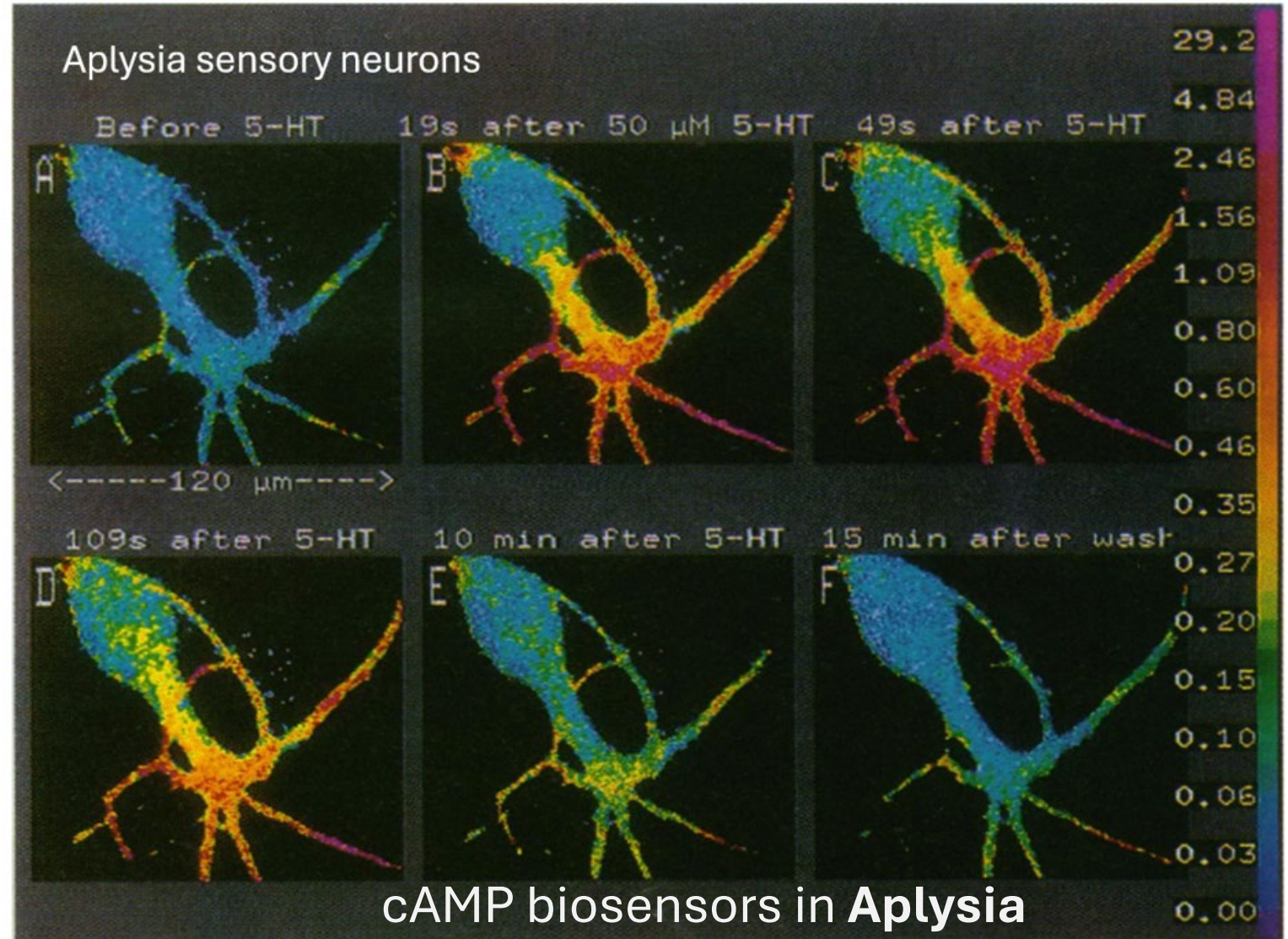
mRNA

growth-stimulating proteins

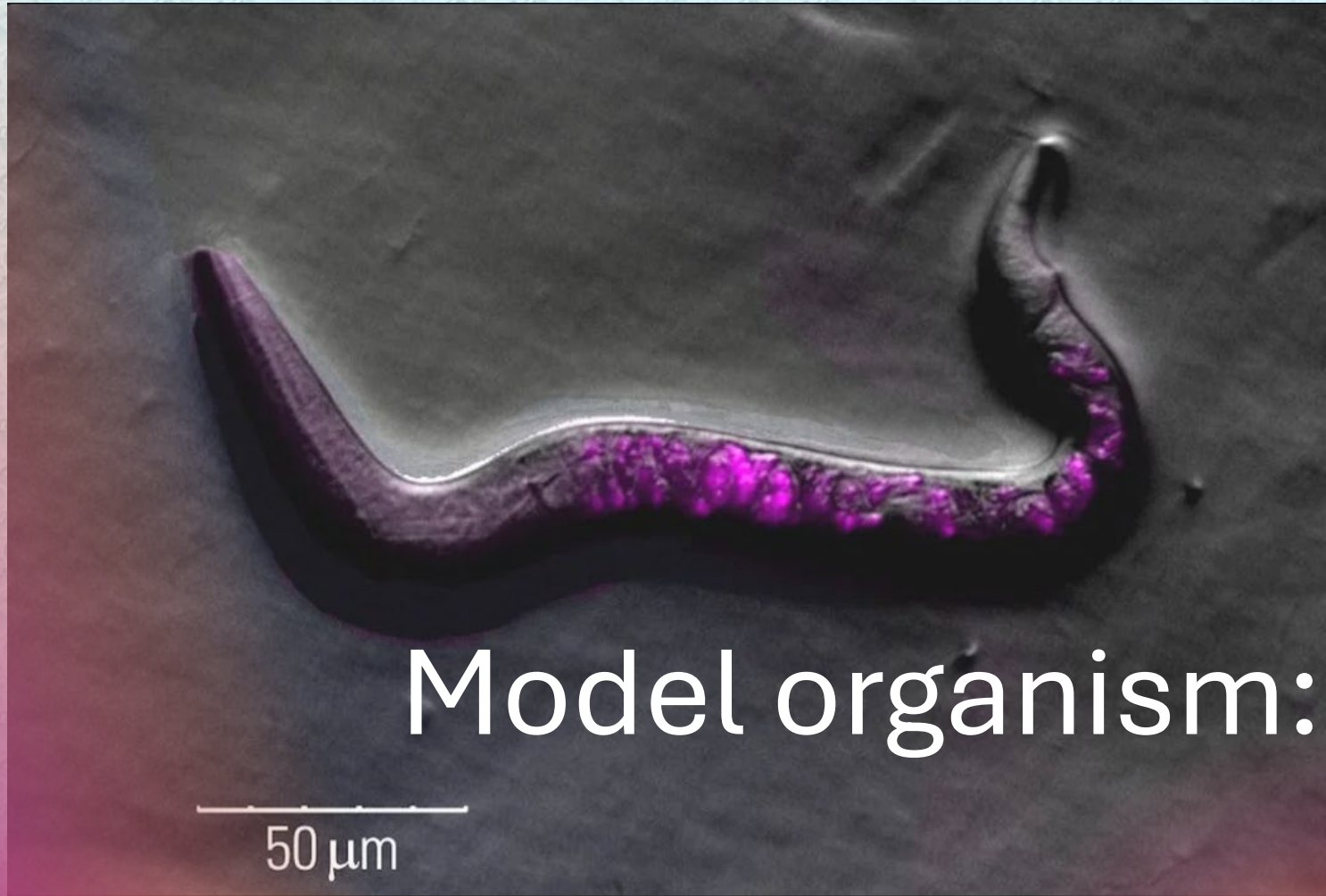
synaptic growth



The **first evidence** of **learning** and **memory**:
Using **sea slugs** as animal models

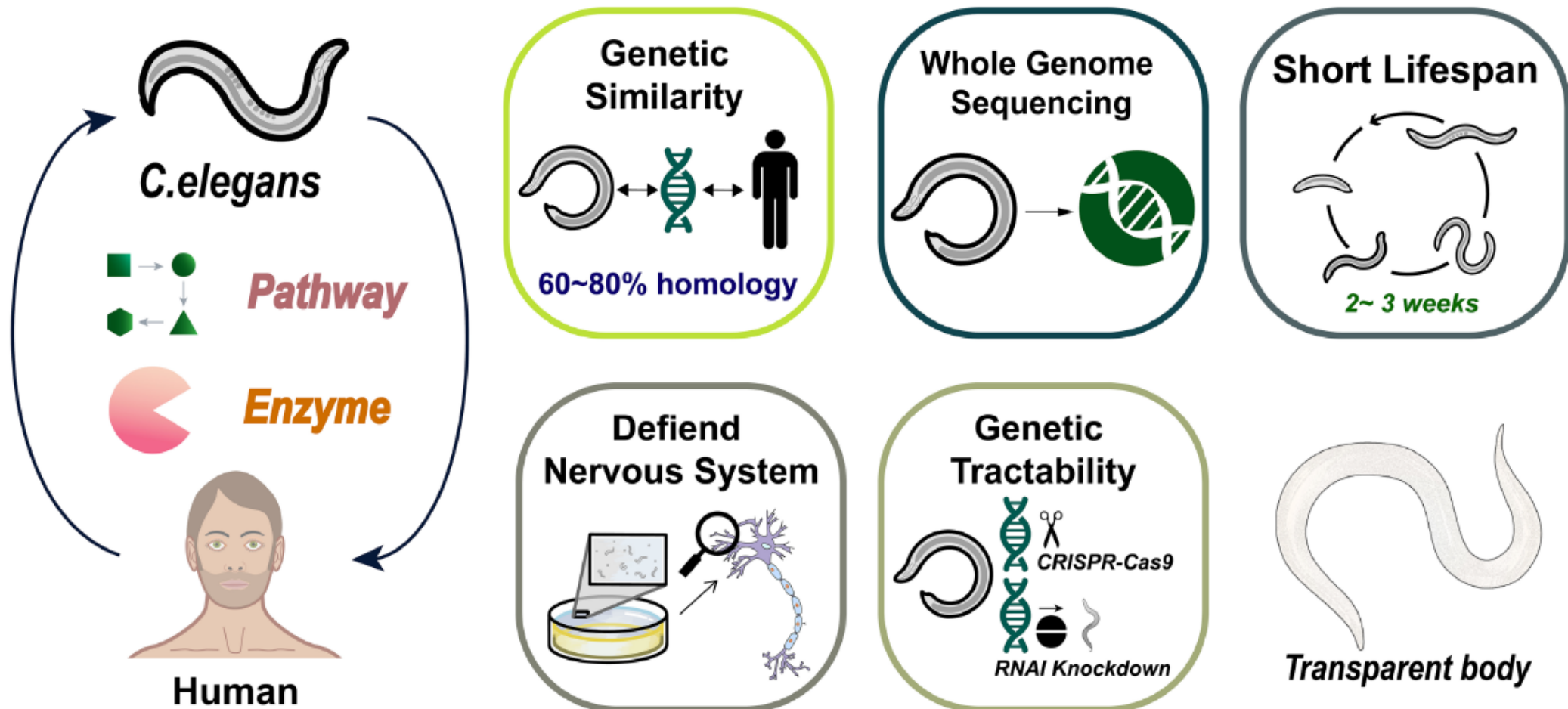


Bacskai et al., 1993 *Science*
From Roger Y. Tsien lab



Model organism: ***C. elegans***

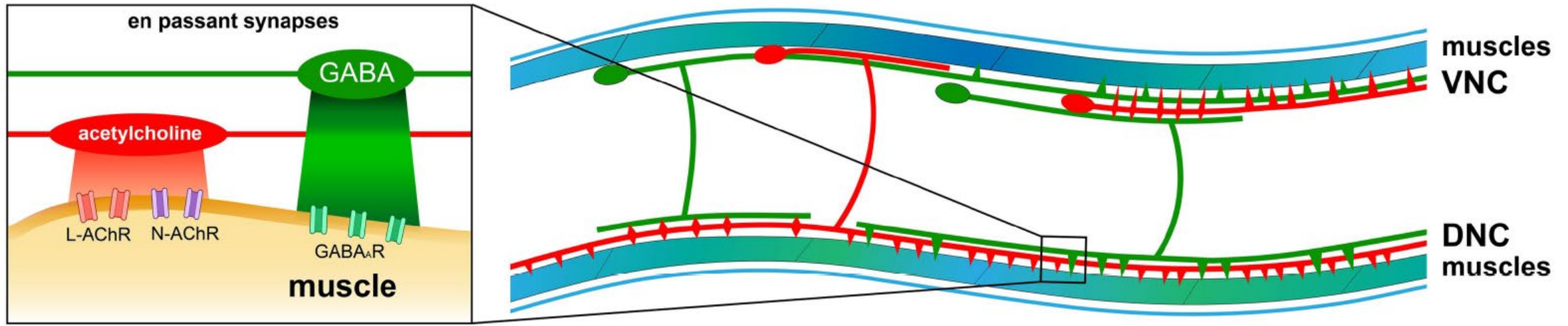
The nematode *C. elegans* as a model organism



Only 302 neurons: Established Connectome

Modified from Pan et al., 2026. Frontiers in Aging Neuroscience

Using *C. elegans* as animal models: motor neurons

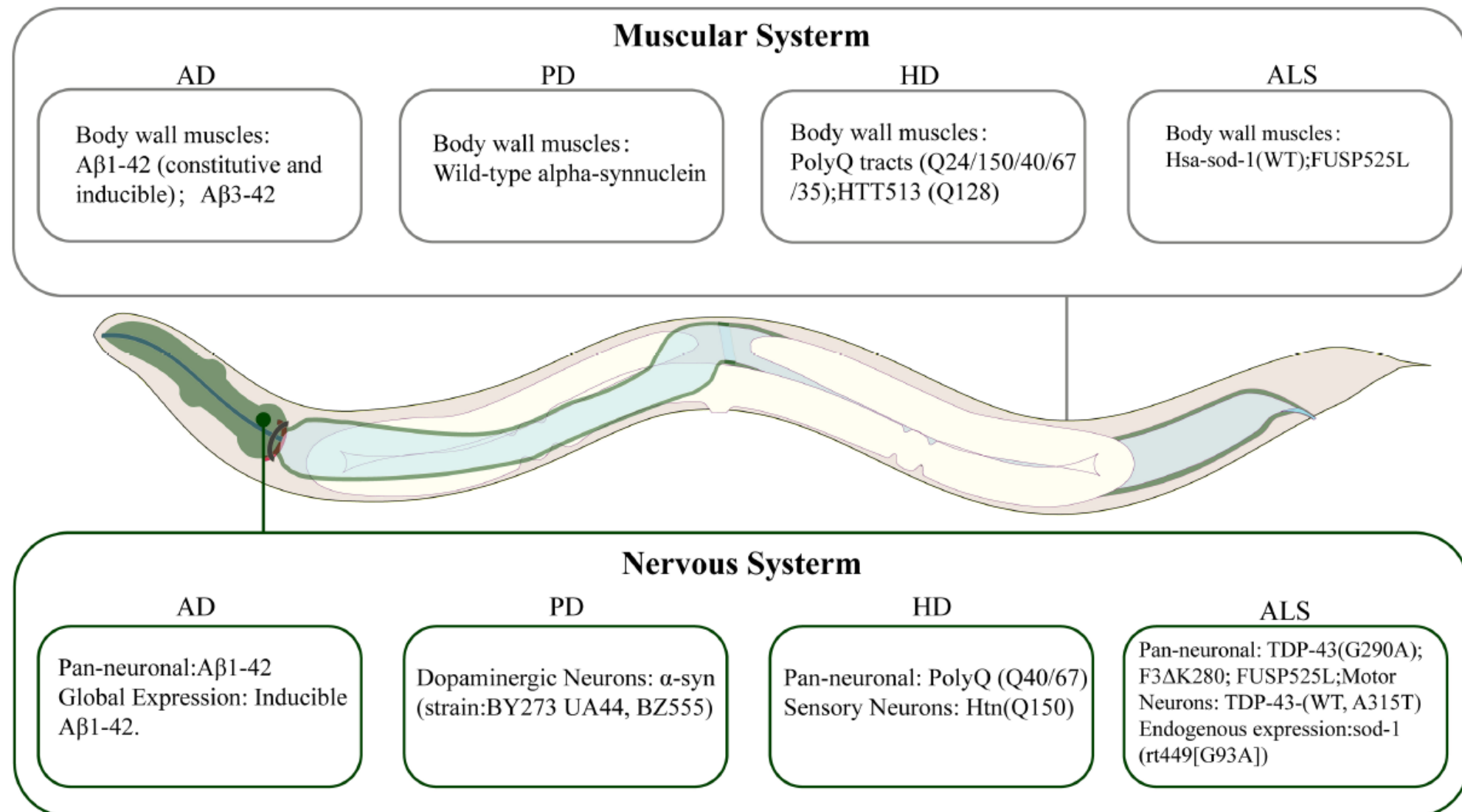


VNC, ventral nerve cord;

DNC, dorsal nerve cord;

GABA, gamma-aminobutyric acid; S/L, short/long (isoforms); N: nicotinic

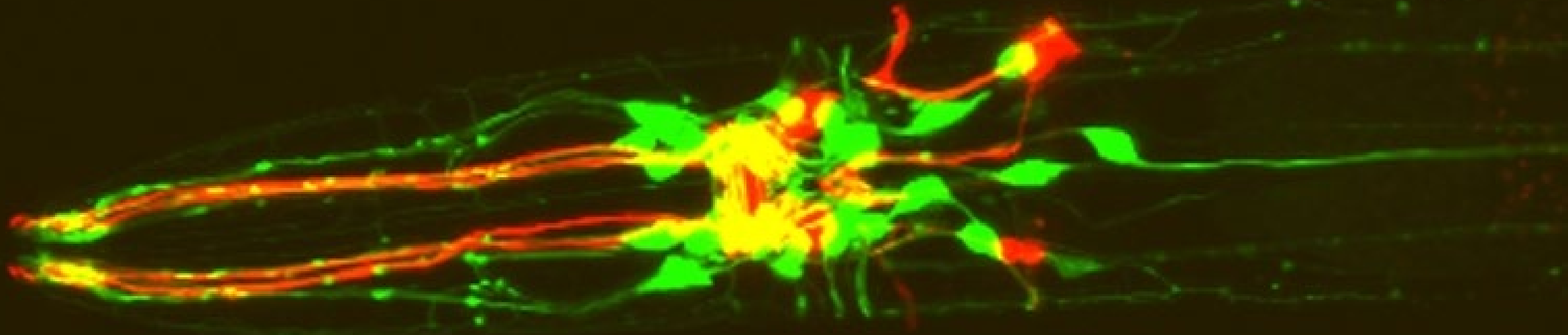
The advantages of using *C. elegans* to model neurodegenerative diseases: **transparent** and **genome accessibility**



The advantages of using *C. elegans* to model neurodegenerative diseases: **transparent** and **genome accessibility**

dopamine neurons (**RFP**)

dopamine receptor-expressing neurons (**GFP**)



Credits:

Image courtesy of Niels Ringstad/MIT

Sensory modality

Main sensory neurons

Sensory neurons in *C. elegans*

O₂ (oxygen)

URX, AQR, PQR

Dominant by URX; AQR and PQR contribute to aerotaxis, (oxygen-dependent behaviors).

CO₂ (carbon dioxide)

BAG, AFD, ASE

Dominant by BAG. AFD (thermosensory neuron) and ASE (salt sensor) also detect CO₂ and contribute to avoidance behavior.

Salt (NaCl)

ASEL, **ASER**

Bilateral ASE neurons: primary gustatory neurons for water-soluble chemicals and salt.

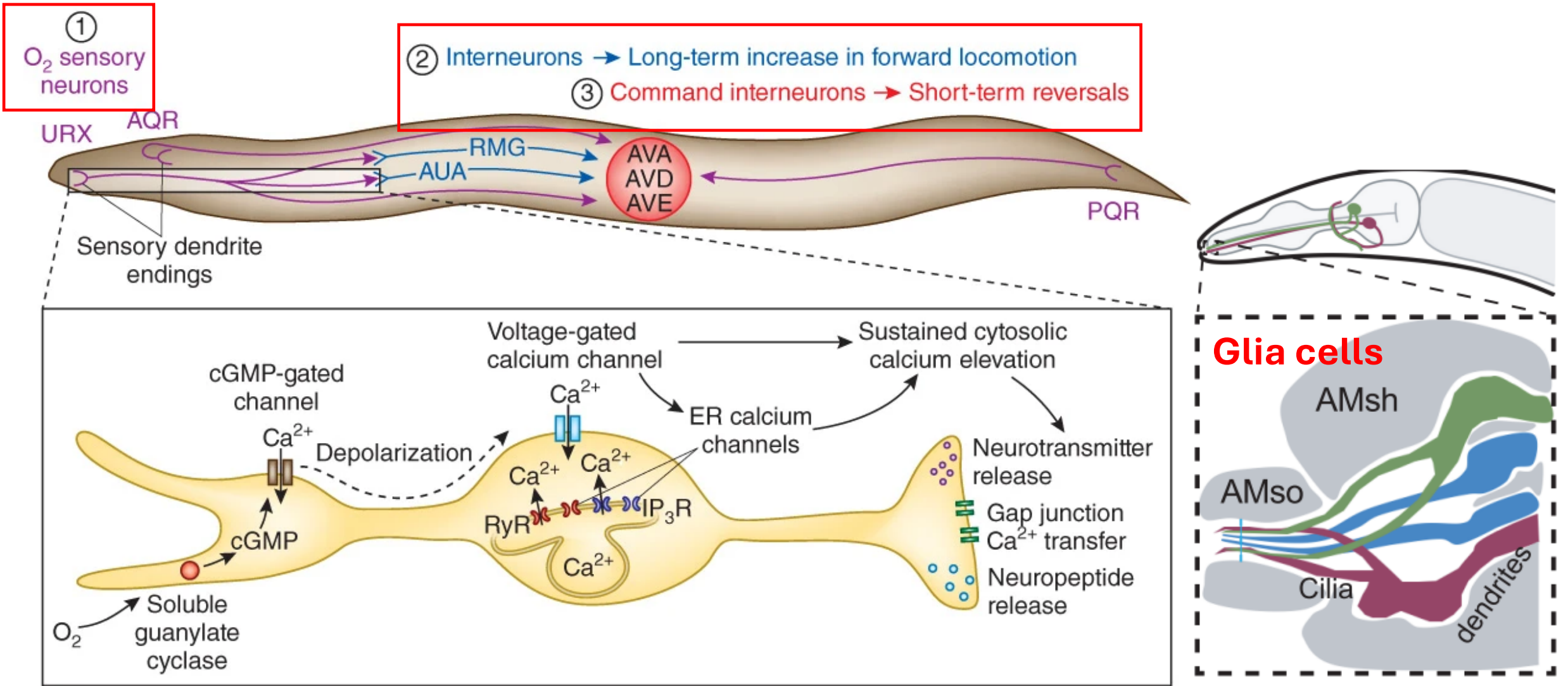
Gentle touch

ALML, **ALMR**, **AVM**,
PLML, **PLMR**, **PVM**

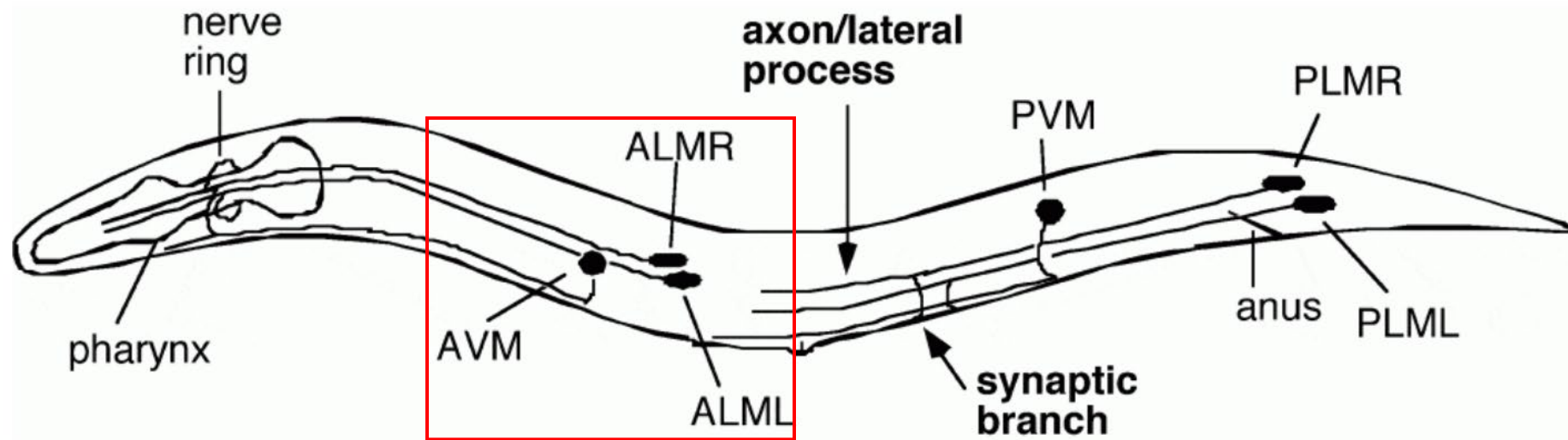
ALM/AVM: anterior touch response (backward movement); PLM: posterior touch response (forward movement).

Serve as animal models for
Sensory Neuropathy, Ciliopathy

Using *C. elegans* as animal models: Sensory neurons (O_2)



Using *C. elegans* as animal models: Sensory neurons (touch)



Nonet Lab, Washington University,
Department of Neuroscience

https://www.youtube.com/watch?v=hTiRT9dK_nU



Model organism: **Zebrafish**

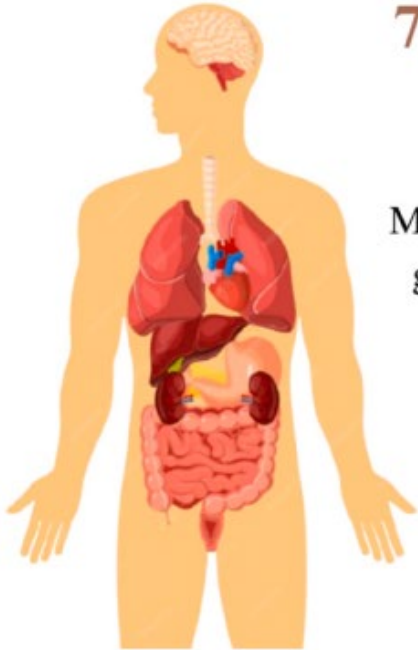
Advantages of using zebrafish as animal models

Human and Zebrafish organ similarities

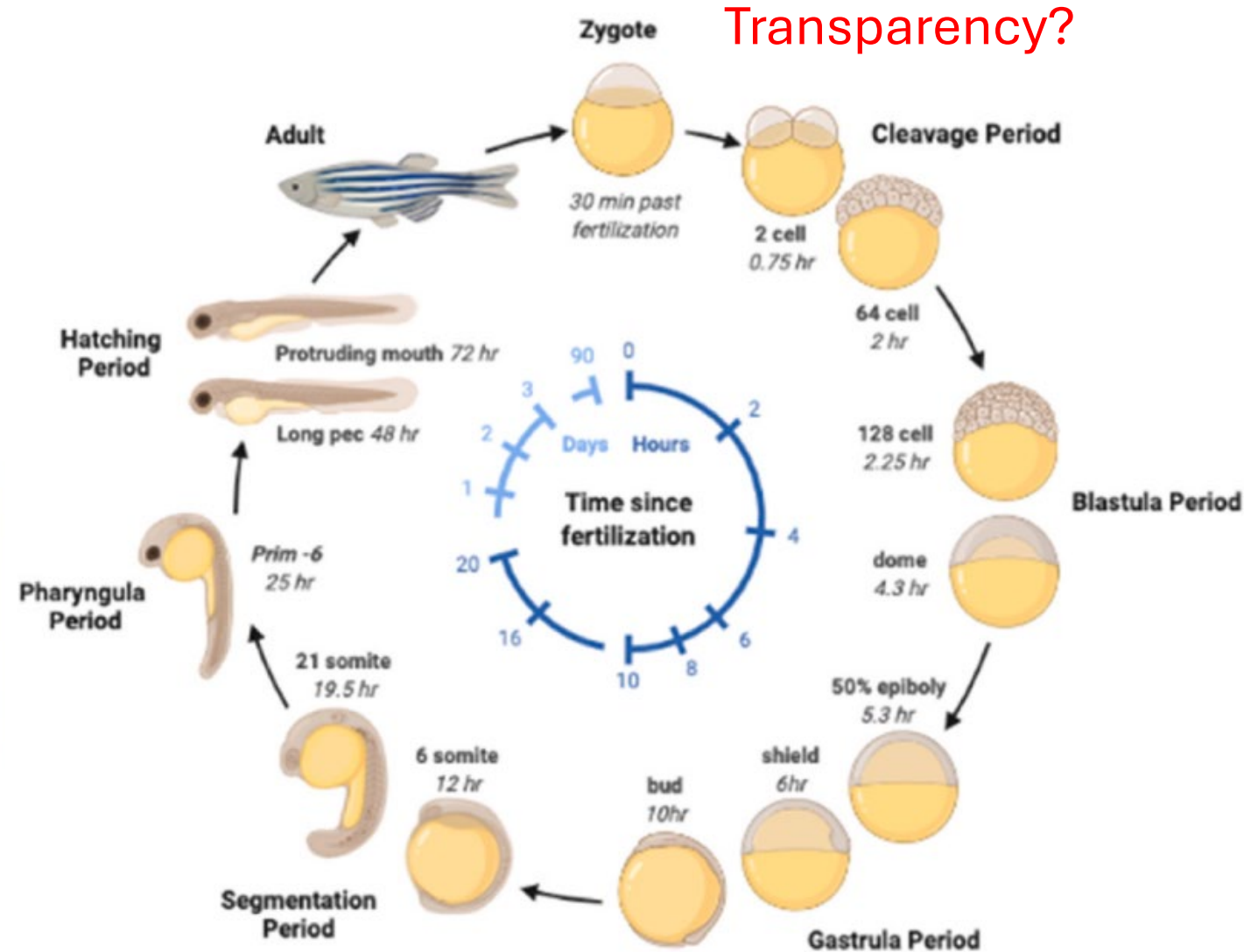
They share many of the same genes and major organ and tissue systems

70% of genes

More than **80%** of genes associated with human disease



- LIVER
- PANCREAS
- GALL BLADDER
- CIRCULATORY SYSTEM
- DIGESTIVE SYSTEM
- LUNGS/SWIM BLADDER
- CENTRAL NERVOUS SYSTEM

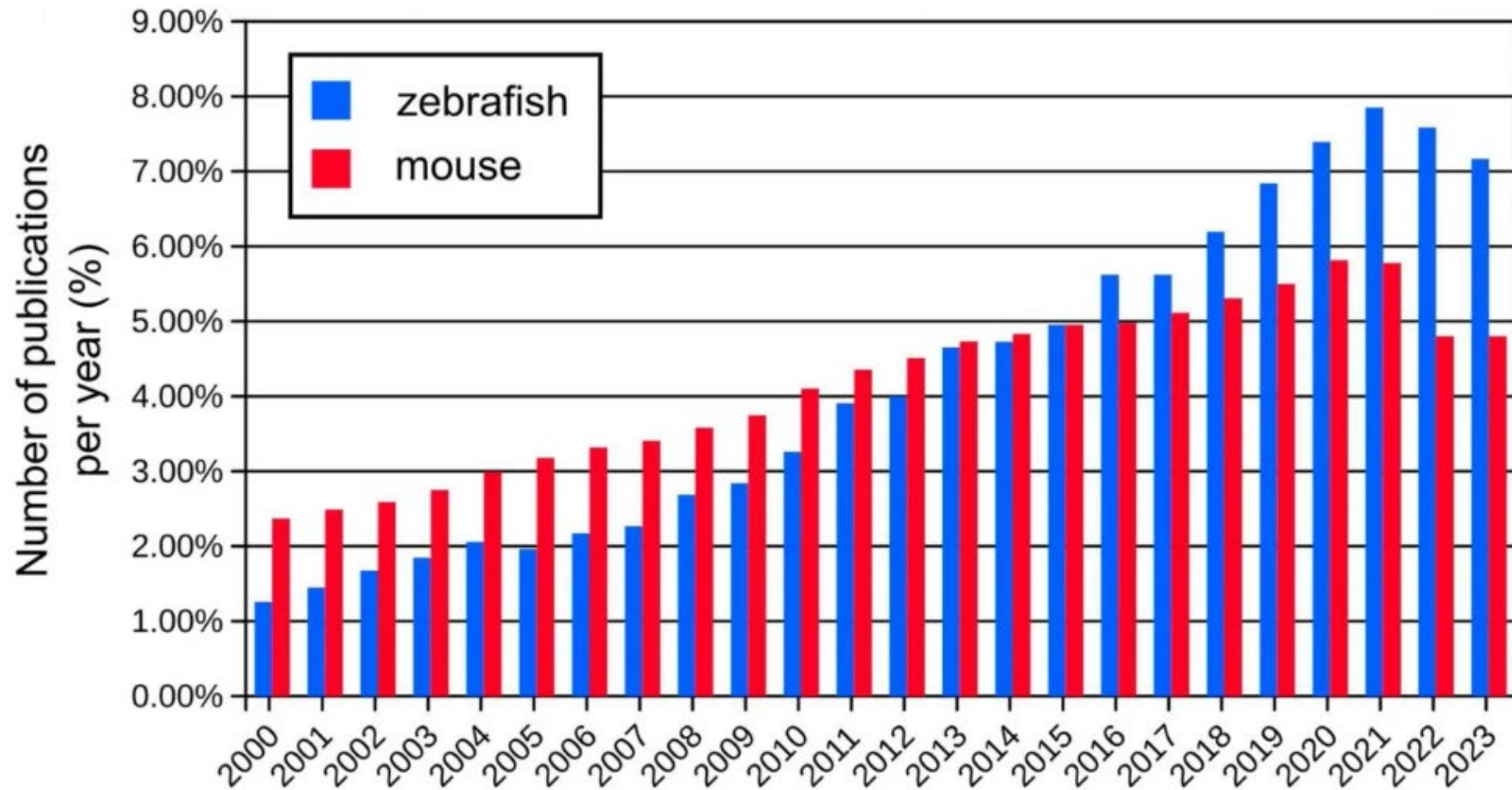


Advantages of using zebrafish as animal models

Short Developmental Period

Transparency (At least before adult)

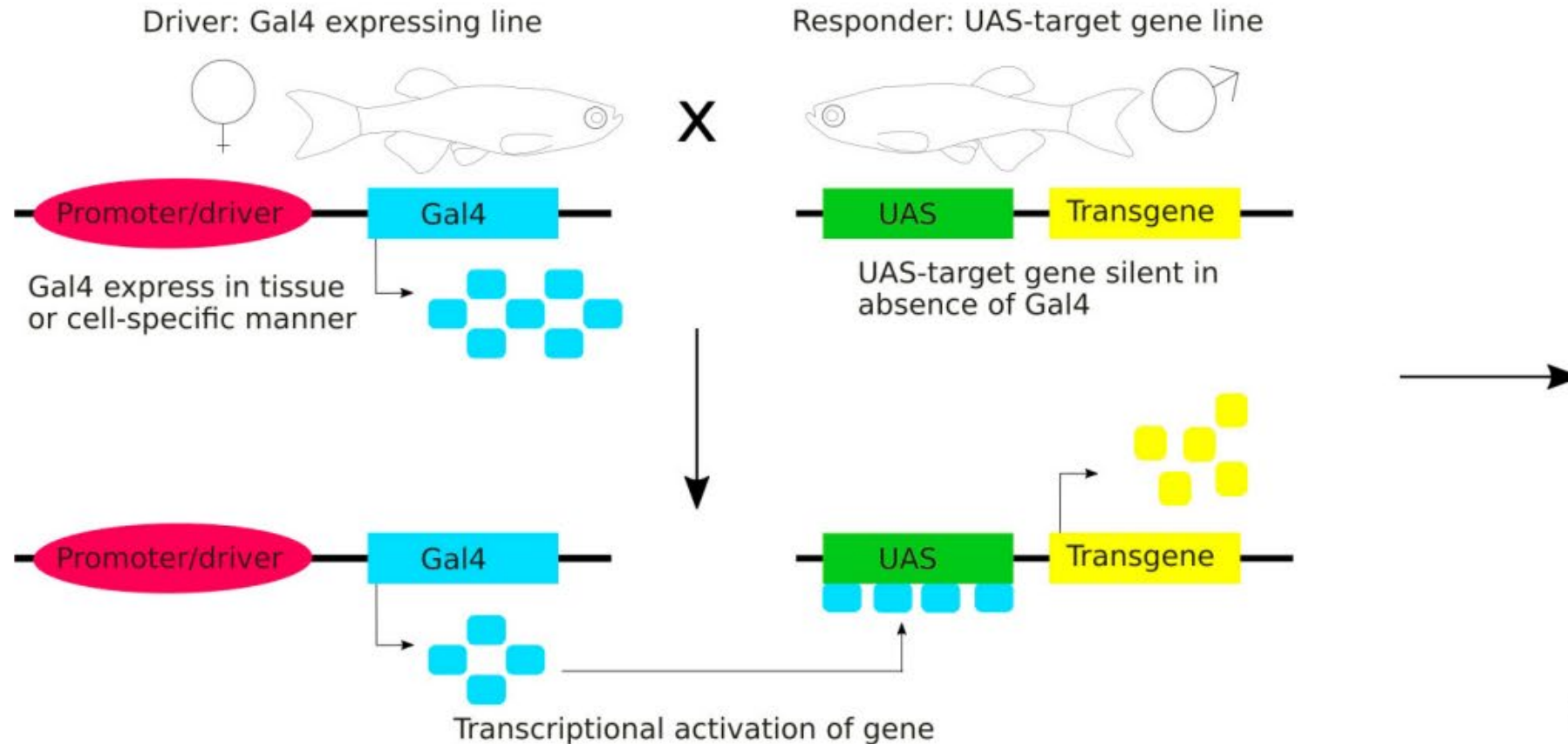
Genetic accessibility



Advantages of using zebrafish as animal models: genetic accessibility

Where?

What?



Generated zc1016c - Gal4
zebrafish line available on zbbrowser.com

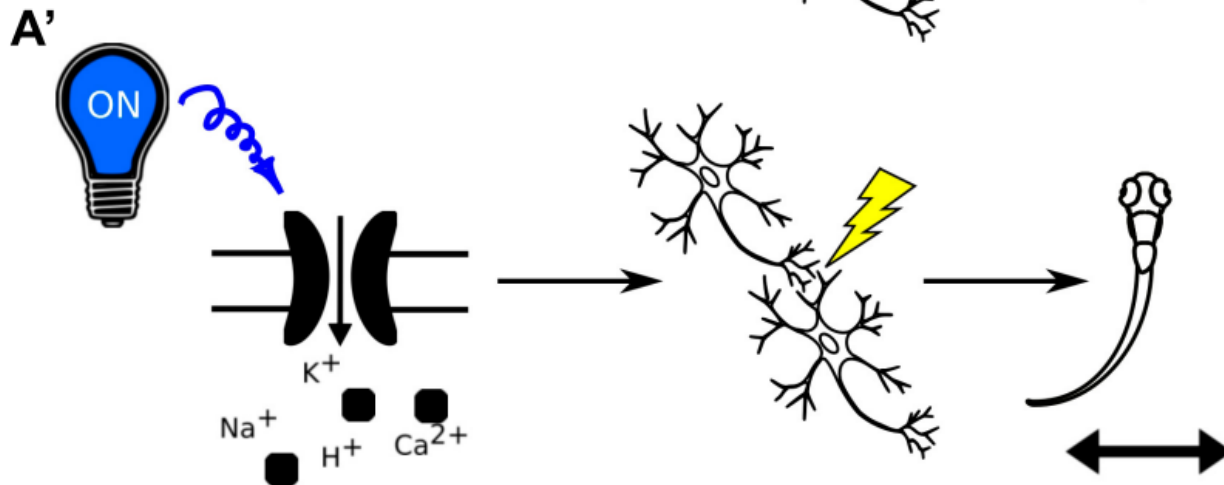
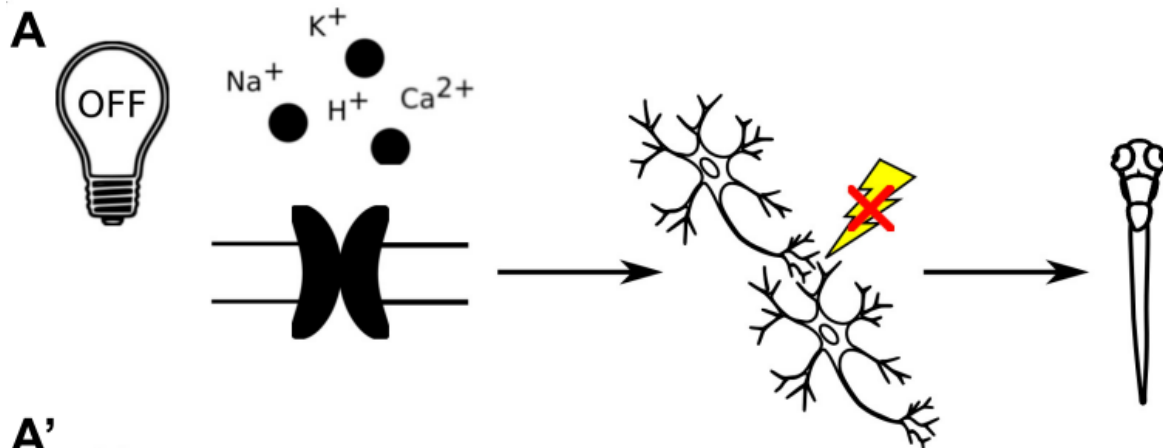
Note: Scientists use similar tools to express genes of interest in *Drosophila*

Advantages of using zebrafish as animal models: transparency

Optogenetics

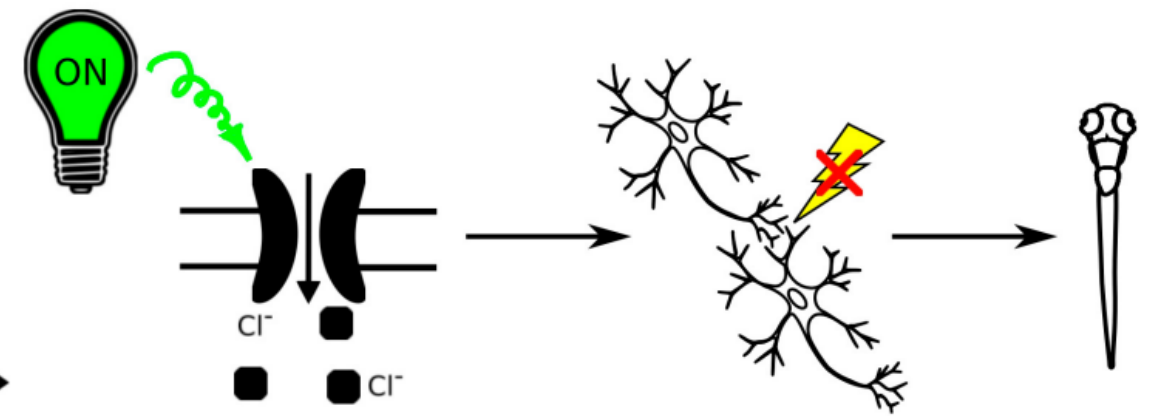
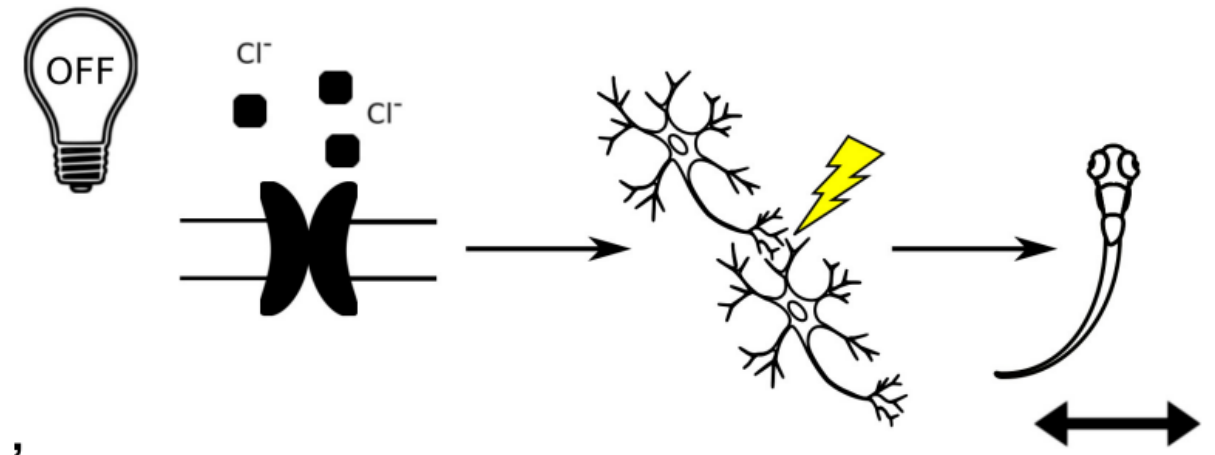
Channelrhodopsins

Derived from algae



Halorhodopsins

Derived from Archaea



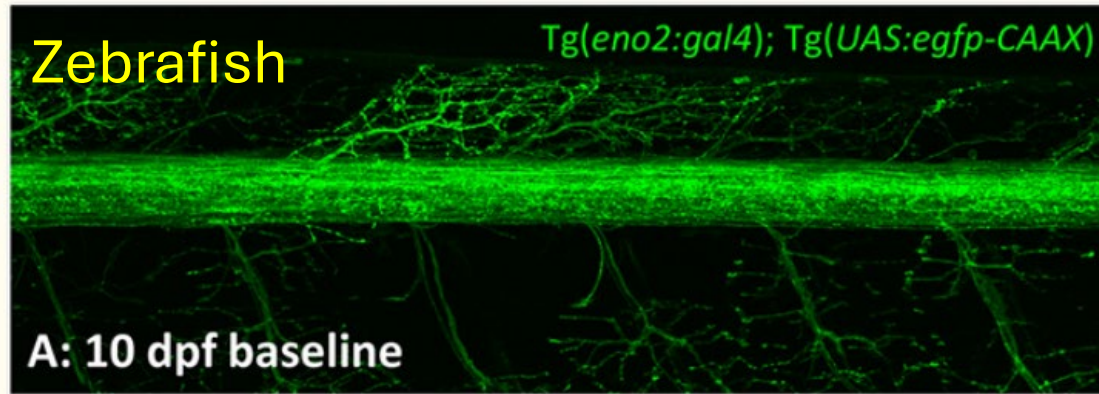
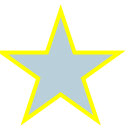


How AI, \$1 billion, and a transparent fish could transform neuroscience

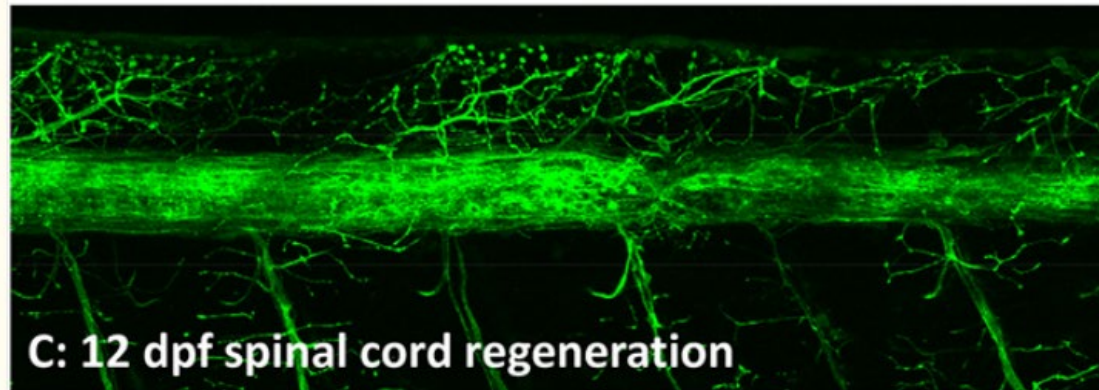
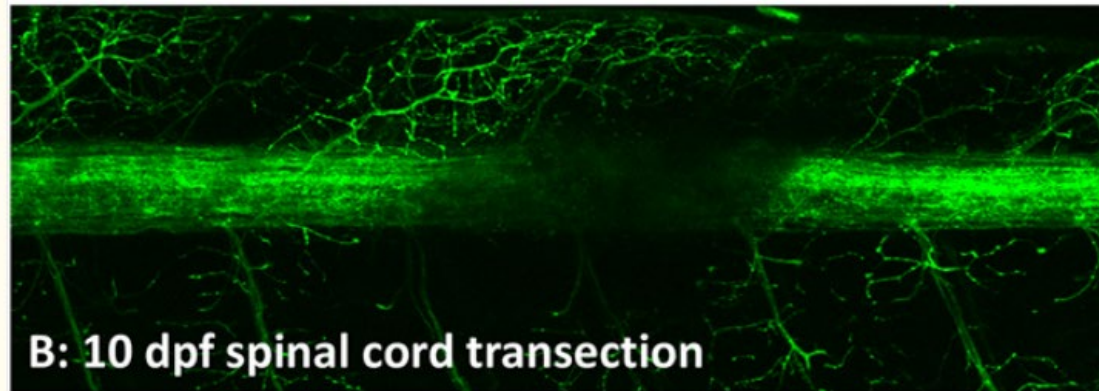
The once-obscure *Danionella* fish takes center stage as a model research organism in a major new project funded by Howard Hughes Medical Institute

23 JUN 2026 • 10:50 AM ET • BY [ZOE BEKETOVA](#)

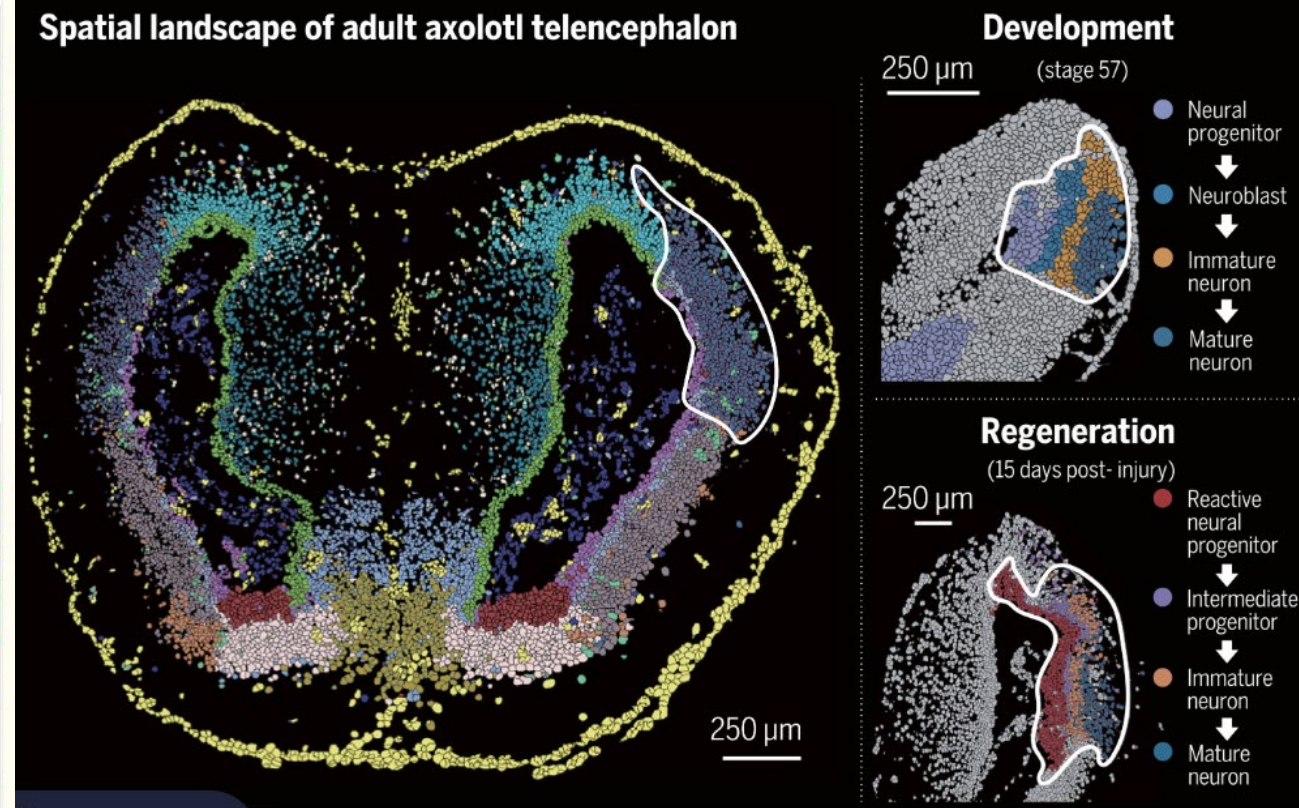
Neurons possess regeneration ability in zebrafish and axolotls



Ambystoma mexicanum



Doszyn et al., 2024 Frontiers in Molecular Neuroscience



Wei et al., 2022 Science



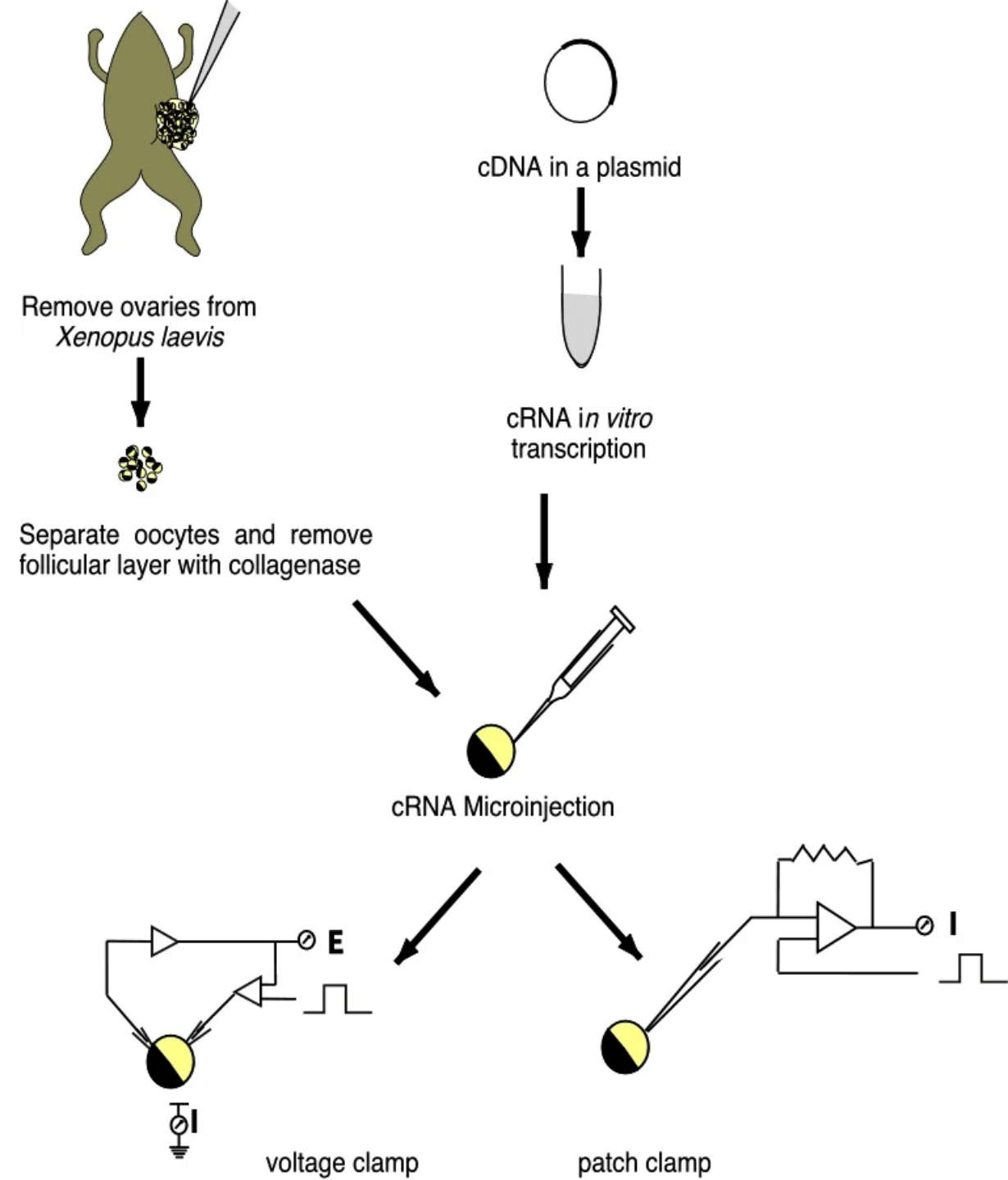
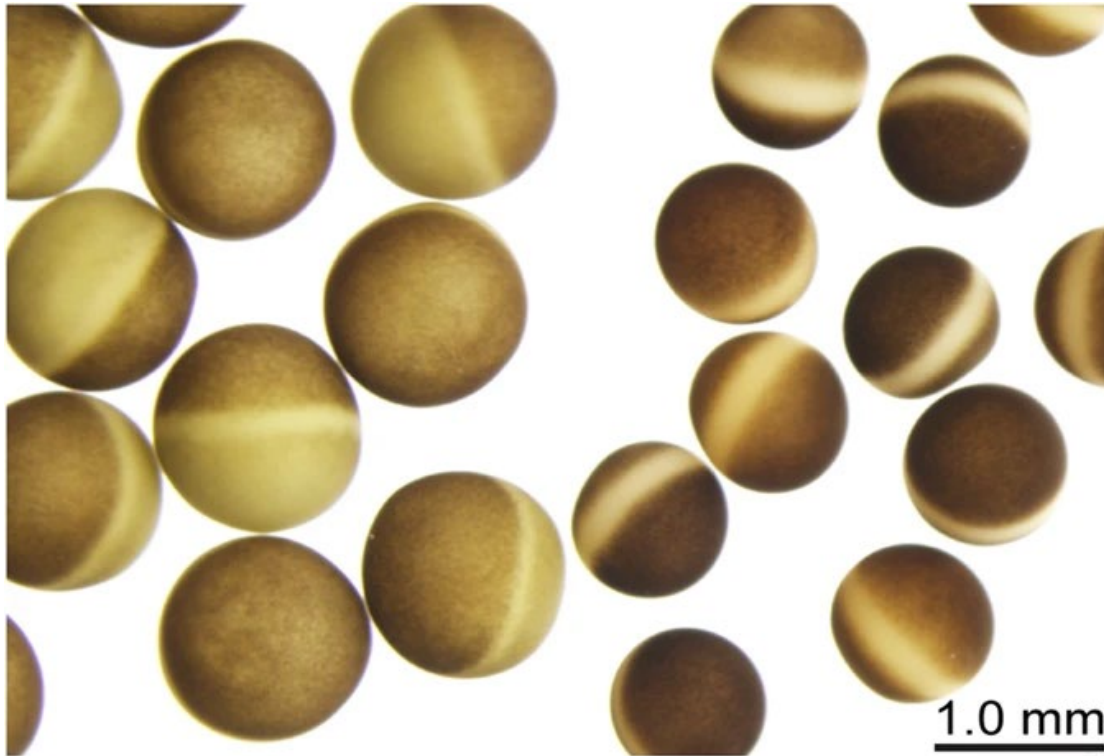
Model organism: **Xenopus**

Zebrafish vs. Xenopus



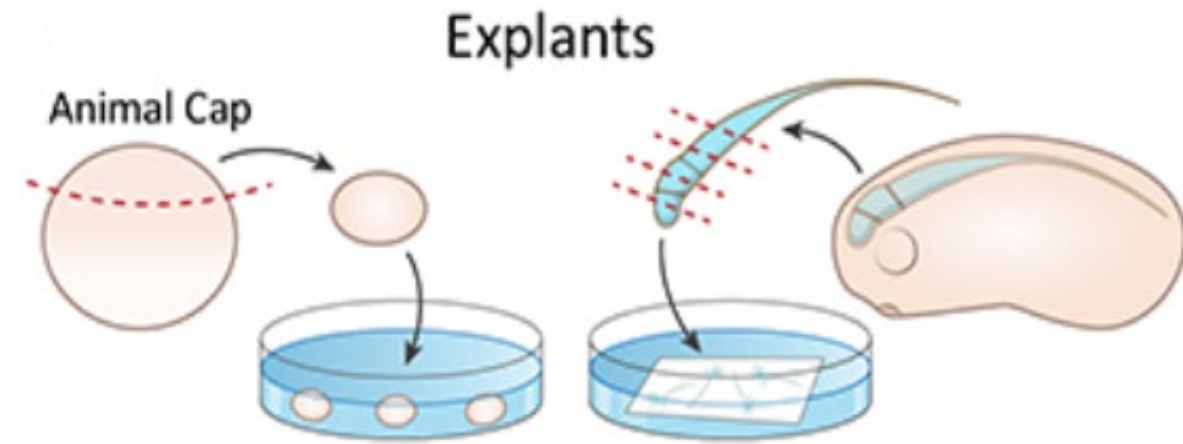
Feature	Zebrafish	Xenopus
Embryo transparency	Excellent. Larvae are highly transparent	Transparent during early stages
Embryological manipulation	Good	Outstanding. Embryos are large and easy to microinject, transplant, and manipulate.
Electrophysiology	Smaller neurons can be challenging.	Large cells facilitate recordings and cell biological studies.
Neural development studies	Excellent	Historically one of the best vertebrate systems for studying early neural induction and patterning.
Cost and space	Low	Moderate
Regeneration studies	Strong CNS regeneration capacity.	Useful for developmental regeneration studies but less used in adult CNS regeneration.

A simple model for studying **ion channel** electrophysiology

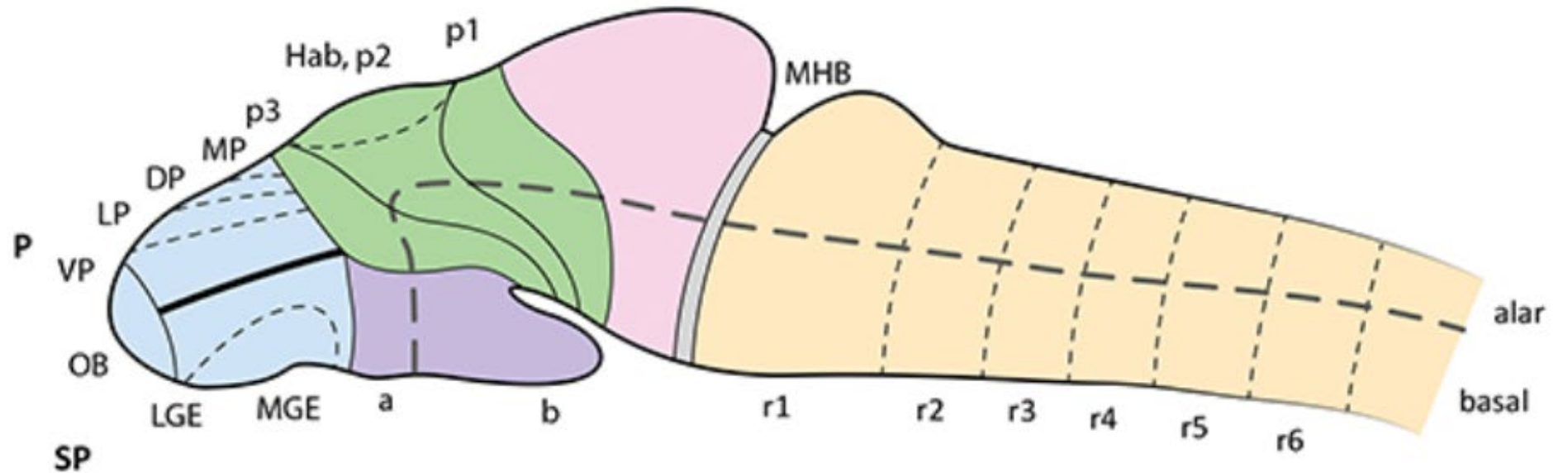


Cristofori-Armstrong et al., 2015. Scientific Reports
Anna Elisabetta Boccaccio (right panel)

A simple model for studying **neurodevelopment**

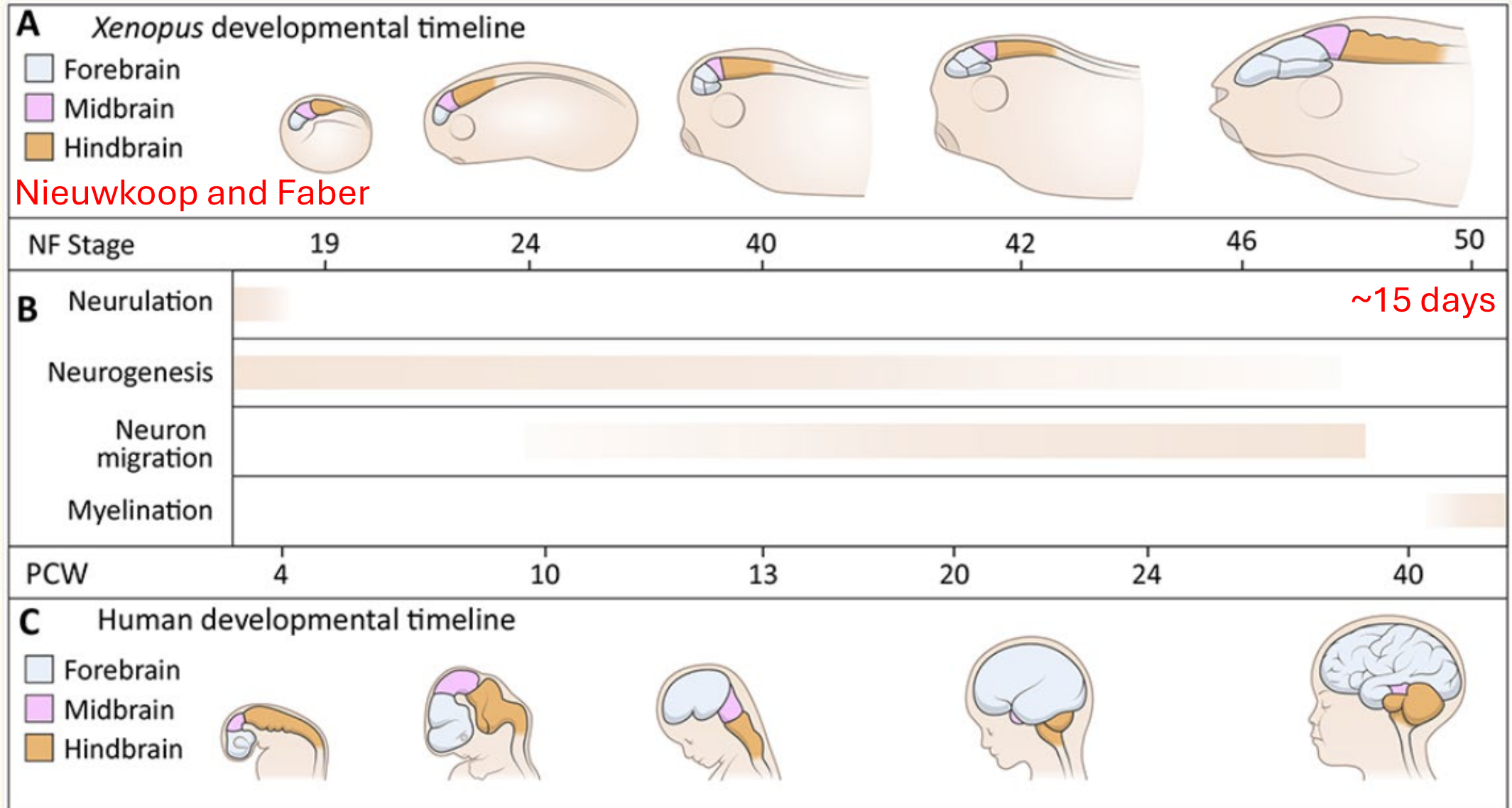


Stage 50



 Telencephalon  Diencephalon  Hypothalamus  Mesencephalon  Rhombencephalon

A simple model for studying **neurodevelopment**



Post-conceptual weeks



Model organism: **Drosophila**



Using Drosophila as models in neuroscience

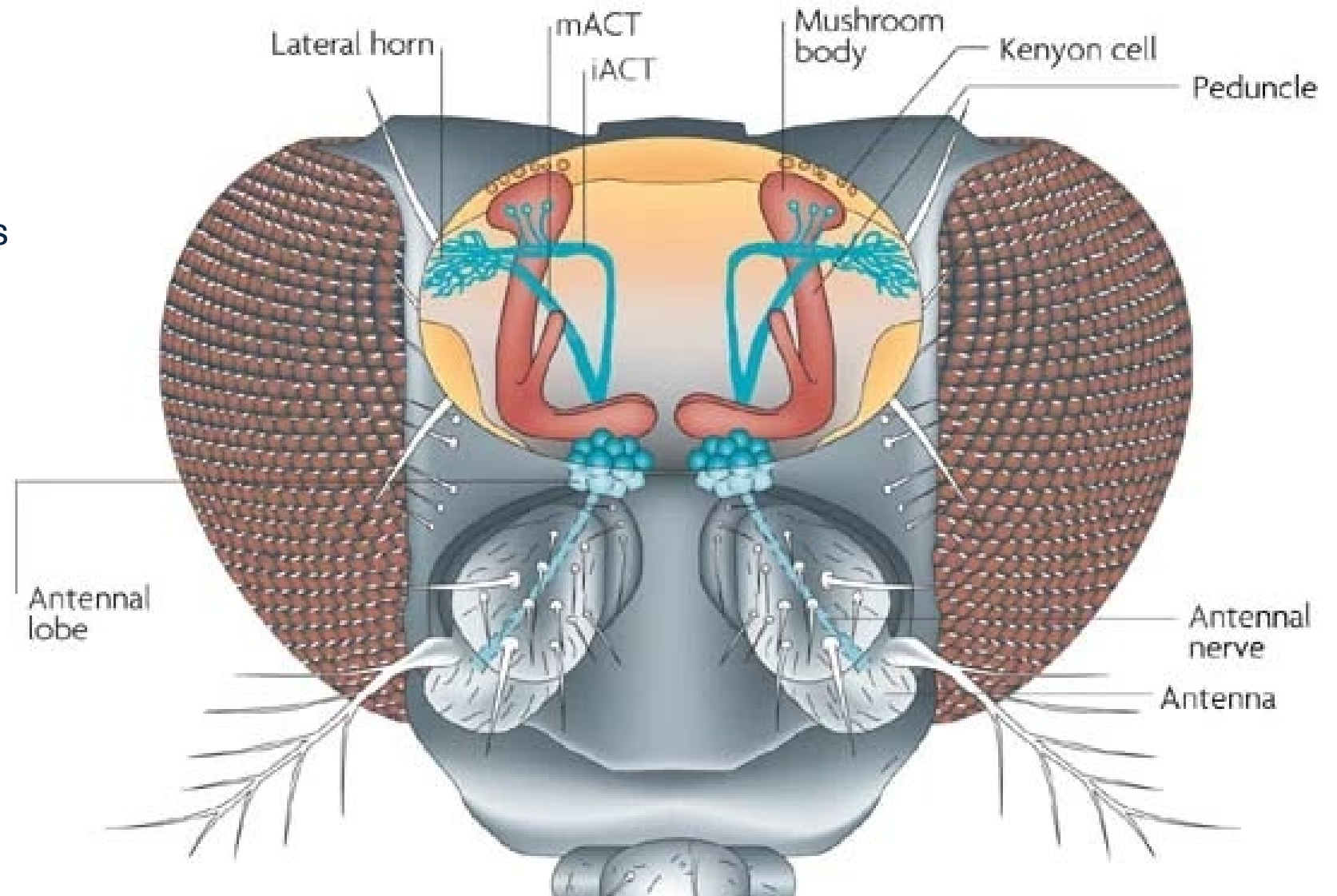


Function	Typical assays
Learning and memory	Olfactory conditioning, courtship memory, long-term memory
Sleep regulation	sleep homeostasis, sleep deprivation and recovery
Sensory processing	Vision, olfaction, taste, hearing, mechanosensation, temperature
Feeding and metabolism	Food choice, hunger, satiety
Reward and motivation	Sugar reward, dopaminergic reinforcement
Social behaviors	Courtship, aggression, group interactions, social learning
Navigation and orientation	Visual navigation, odor tracking, place learning
Motor control	Walking, grooming, flight control, locomotor coordination
Aging and neurodegeneration	Alzheimer's-, Parkinson's-, and Huntington's-like disease models

Using *Drosophila* as models in neuroscience

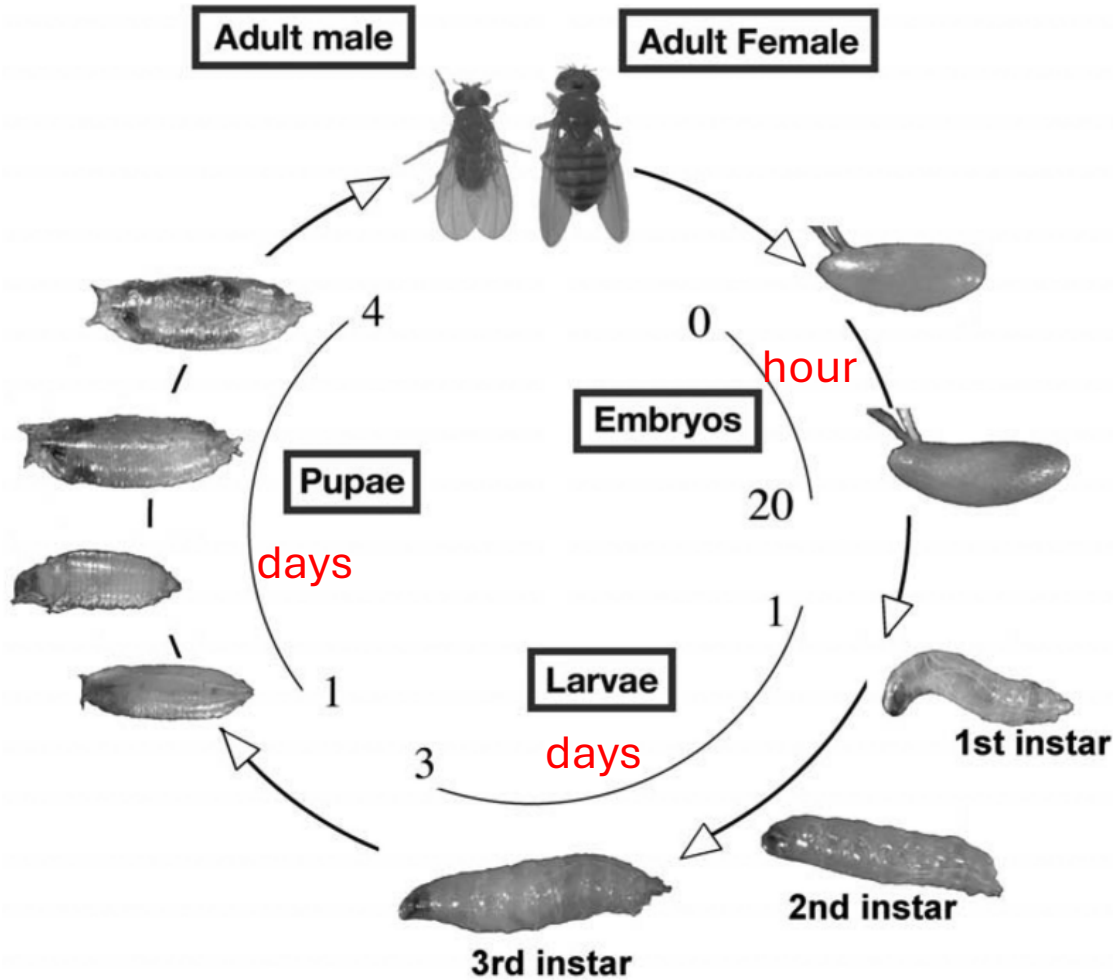
Key facts for the ***Drosophila*** nervous system:

- 140,000 to 166,000 neurons
- Established neural circuitry
- Short maturation time
- Easy genetic manipulation (Gal4-UAS system)



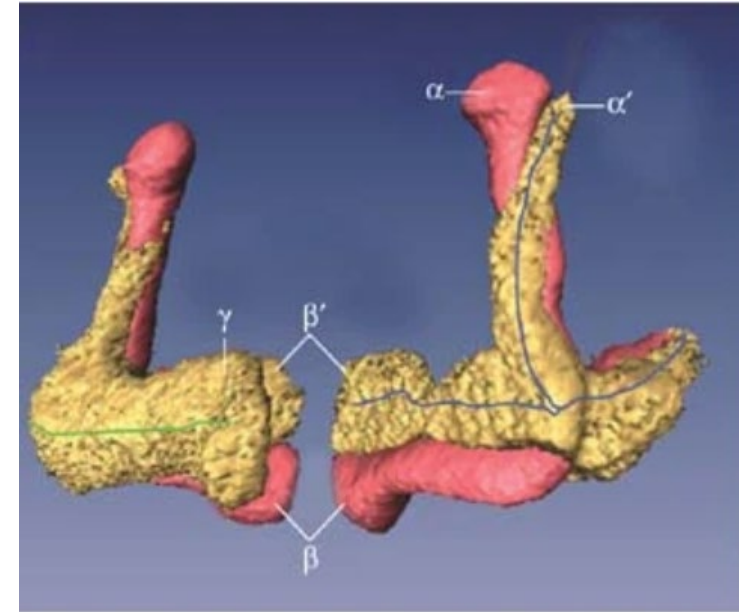
inner and medial antennocerebral tract (iACT and mACT)

- Short maturation time



Fernández-Moreno et al., 2007. *Methods Mol Biol*

- Established neural circuitry



Nature Reviews | Neuroscience

processes sensory information,
coordinates associative
learning, and guides behavioral
decisions

Keene and Waddell (2007) *Nature Reviews Neuroscience*

Model organisms



Marmoset

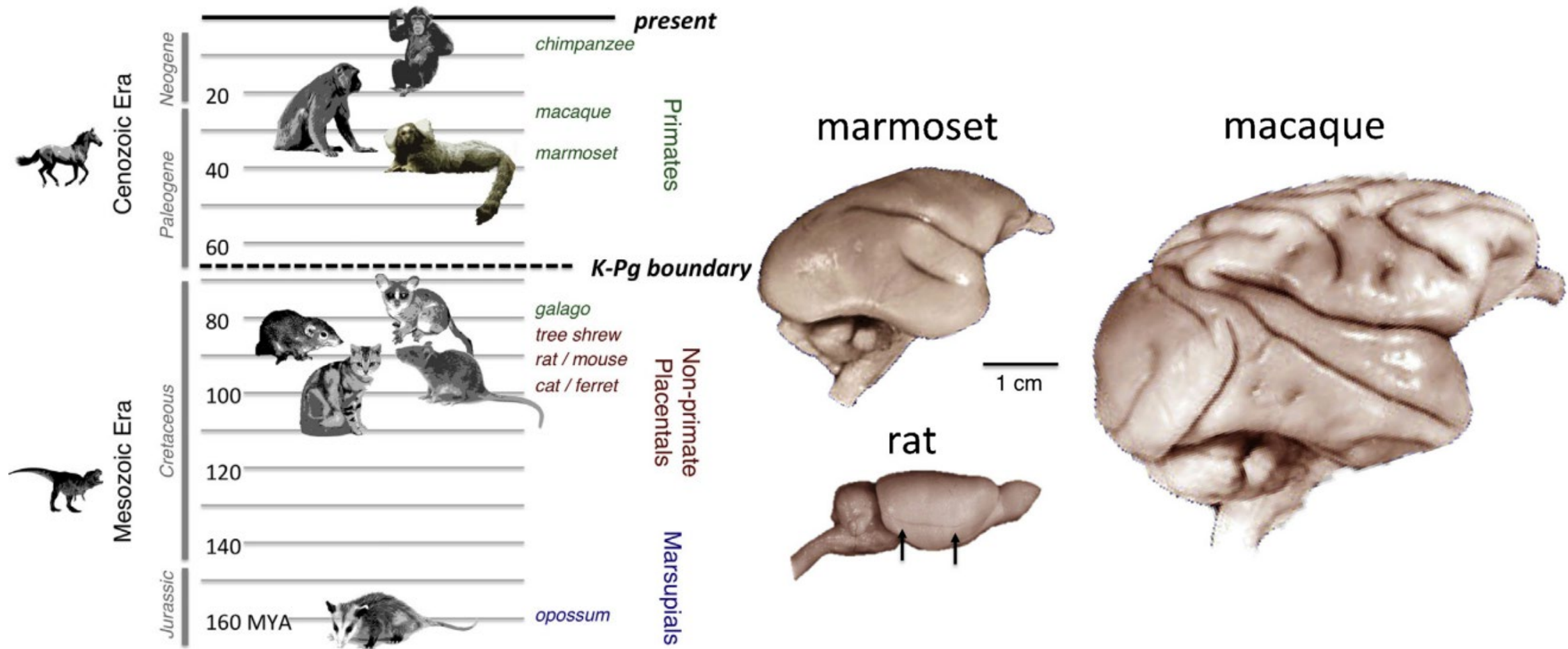


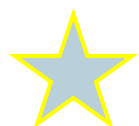
Mouse



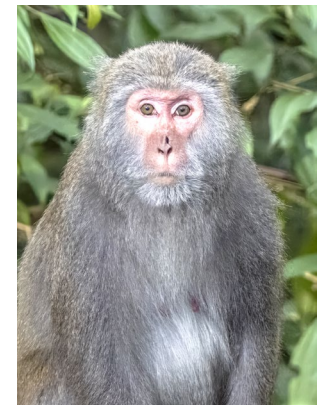
Macaque

Comparisons between mouse (rat), marmoset, and macaque





Feature	Mouse	Marmoset	Macaque
Brain size	~0.4-0.5 g	~8-10 g	~80-100 g
Cortical organization	Smooth (lissencephalic)	smooth cortex with some expansion	Highly folded (gyrencephalic) cortex
Prefrontal cortex development	Limited	Moderately developed	Highly developed
Genetic manipulation	Excellent; extensive transgenic and knockout tools	Growing toolkit; transgenic lines available	Difficult and expensive, though CRISPR



Feature	Mouse	Marmoset	Macaque
Breeding speed	Very rapid (2-3 months to adulthood)	Moderate (12-18 months maturation)	Slow (3-5 years maturation)
Lifespan	~2-3 years	~10-15 years	~25-40 years
Housing requirements	Simple	Moderate	Complex and expensive
In vivo imaging accessibility	Excellent; genetically encoded indicators widely used	Good; increasingly used for imaging	Good but technically demanding
Electrophysiology	Excellent for circuit studies	Excellent	Gold standard for neurophysiology

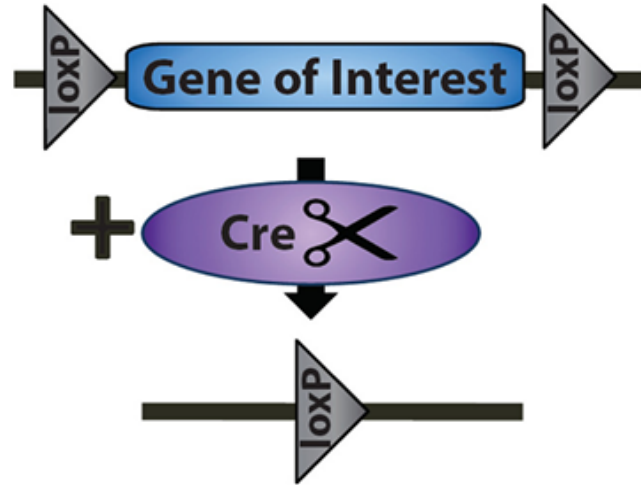


Feature	Mouse	Marmoset	Macaque
Behavioral complexity	Basic to moderate	High social complexity	Very high cognitive flexibility
Visual system similarity to humans	Relatively poor	Good	Excellent
Motor system similarity to humans	Moderate	Good	Excellent
Language-related research	Limited	Moderate (vocal communication studies)	Strong for higher cognitive and communication studies
Disease modeling	Widely used for genetics and neurodegeneration	Emerging for psychiatric and developmental disorders	Strong for cognitive, psychiatric, and translational studies

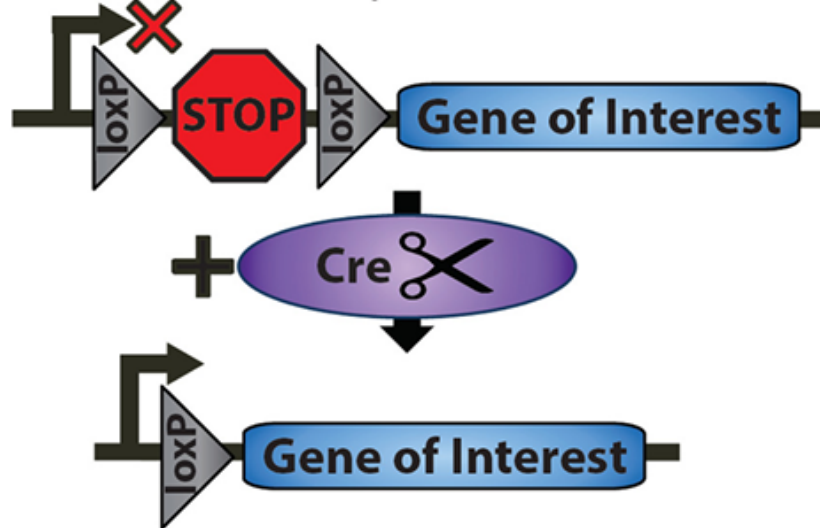


Common mouse models for brain research

A. Cre-induced deletion



B. Cre-induced expression



(A) Knockout models:

- Study the functions of a specific gene
- Disease modeling

(B) Knock-in models:

- Express a humanized gene
- Express a mutant gene (dominant negative)

(C) Mutagenesis:

- Express a human mutant gene
(2020 Nobel Prize: CRISPR Cas9 technology)



Emmanuelle Charpentier
and Jennifer Doudna

iPSC for brain research

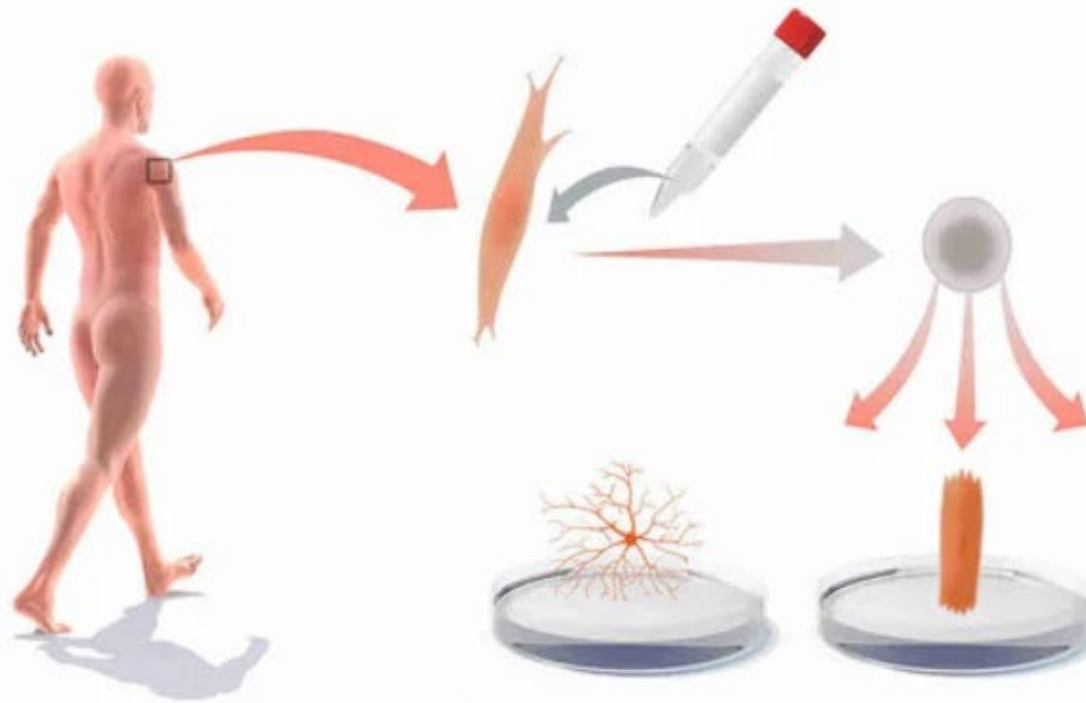
2012 Nobel Prize of Physiology or Medicine



Shinya Yamanaka
University of Kyoto, Japan
Photo Credit:
Center for iPS cell Research and Application, Kyoto University



John B. Gurdon
Gurdon Institute in Cambridge, UK



iPS cells can now be generated from humans, including patients with disease. Mature cells including nerve, heart and liver cells can be derived from these iPS cells, thereby allowing scientists to study disease mechanisms in new ways.

© 2012 The Nobel Committee for Physiology or Medicine
The Nobel Prize® and the Nobel Prize® medal design mark are registered trademarks of the Nobel Foundation

Illustration and layout: Mattias Karlén

iPSC vs. transgenic mouse models for brain research

Feature	Human iPSC Model	Mouse Model
Species relevance	authentic human genetics	Mouse genome
Patient-specific mutations	Easily generated from individual patients	Requires genetic engineering
Cell types studied	Neurons, astrocytes, oligodendrocytes, microglia, organoids	All brain cell types in vivo
Brain circuitry	Limited; organoids have partial networks	Full neural circuits and behavior
Behavioral analysis	Not possible	Major advantage
Drug screening	High-throughput and human-specific	Lower throughput and more expensive
Developmental disorders	Excellent for studying human neurodevelopment	Some human-specific features absent
Cost and time	Moderate; months	Higher; can require years
Translation to patients	Often higher cellular relevance	Frequently limited by species differences

Developmental xenocortication using human-derived organoids in mice

<https://doi.org/10.1038/s41586-026-11032-2>

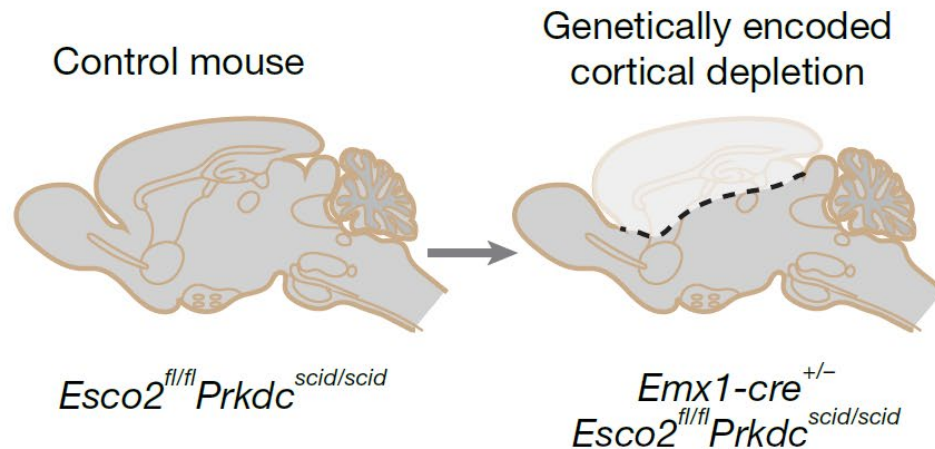
Received: 27 June 2025

Accepted: 12 August 2026

Published online: 16 September 2026

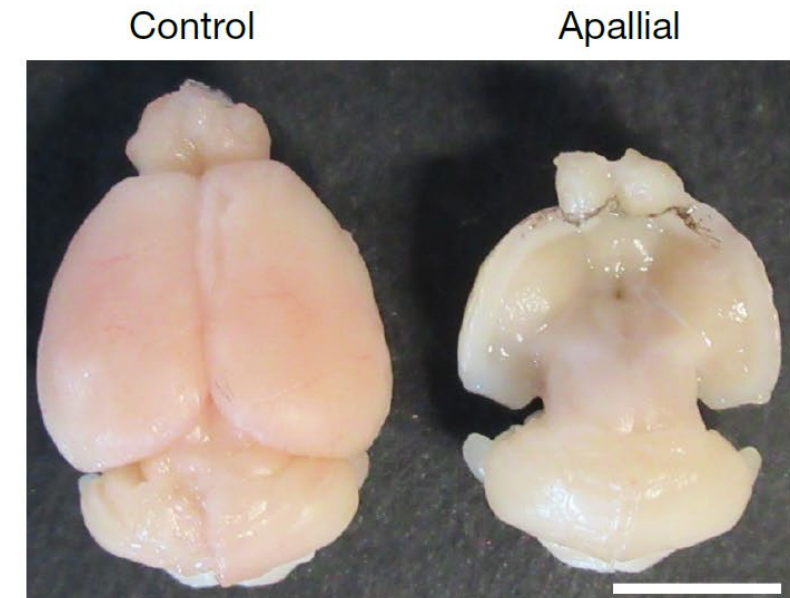
Konstantin Kaganovsky^{1,2,9}, Kevin W. Kelley^{1,2,9}, Tilo Gschwind^{3,9}, Paul M. Harary^{1,2,9}, John Kochalka¹, Alexander D. White⁴, Garikoitz Lerma-Usabiaga^{5,6}, Xiaoyu Chen^{1,2}, Omer Revah^{1,2}, Felicity Gore^{1,4}, Ayano Aoyama¹, Jennifer L. Shadrach³, Se-Jin Yoon^{1,2}, Alfredo Valencia^{1,2}, Satoe Ogawa^{1,2}, Noah Reis^{1,2}, Hannes Vogel⁷, Brian Wandell⁸, Julia A. Kaltschmidt³, Ivan Soltesz³, Karl Deisseroth^{1,4} & Sergiu P. Pasca^{1,2}✉

a



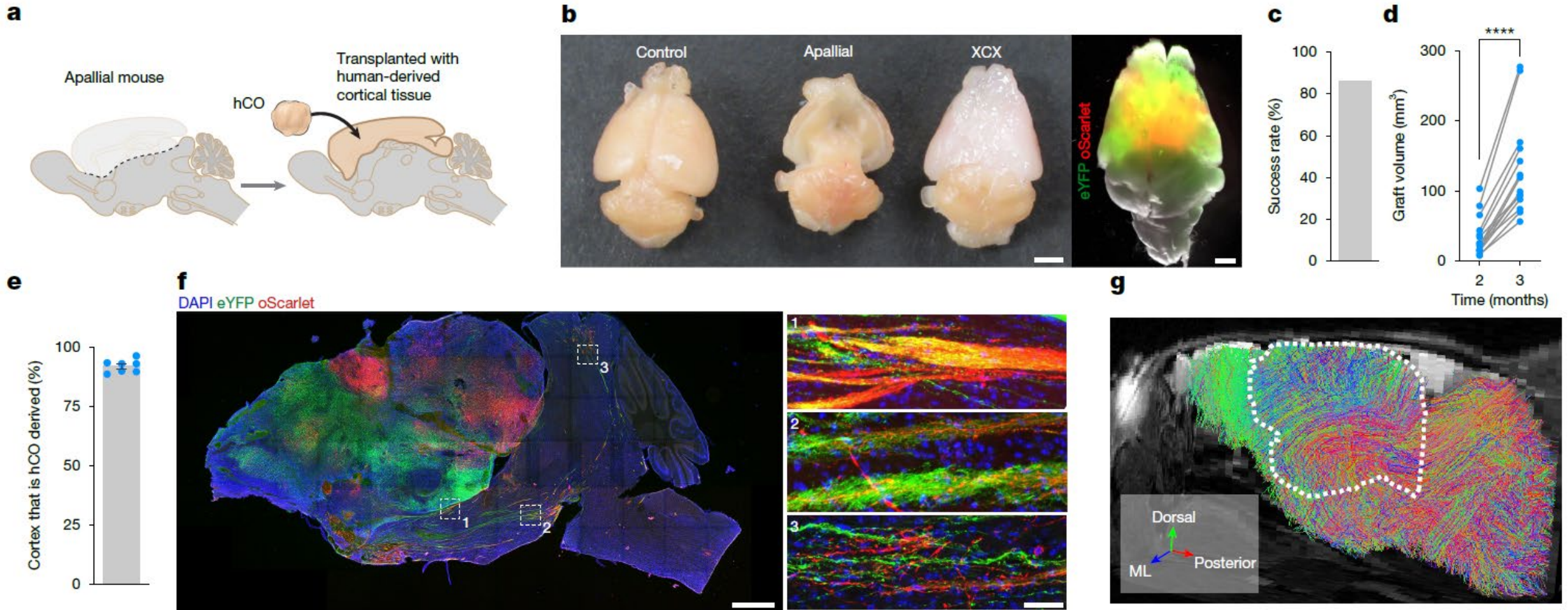
a (non) + pallial (mantle) → apallial

b



Kaganovsky et al., 2026. *Nature*

The mouse model with a human brain

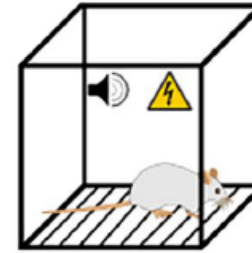
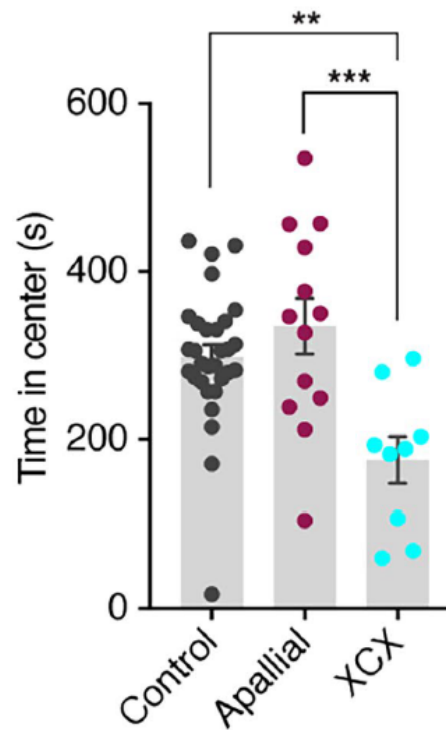
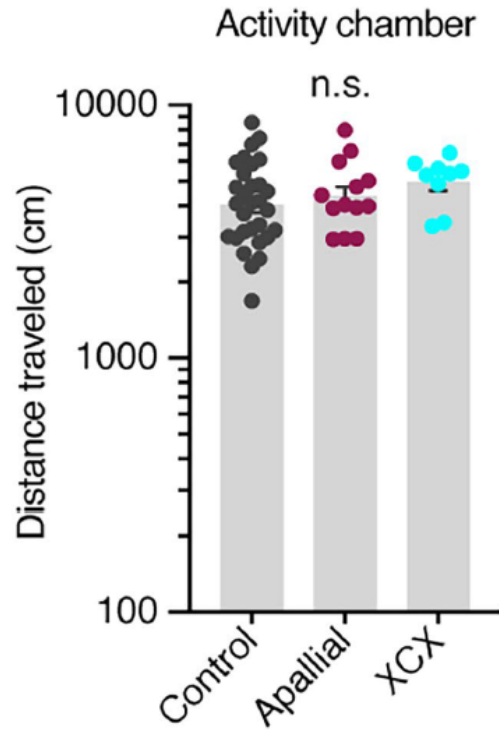
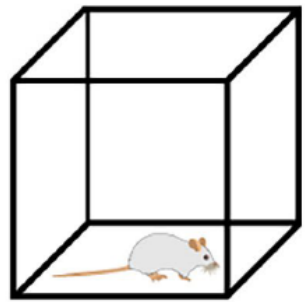


Kaganovsky et al., 2026. *Nature*

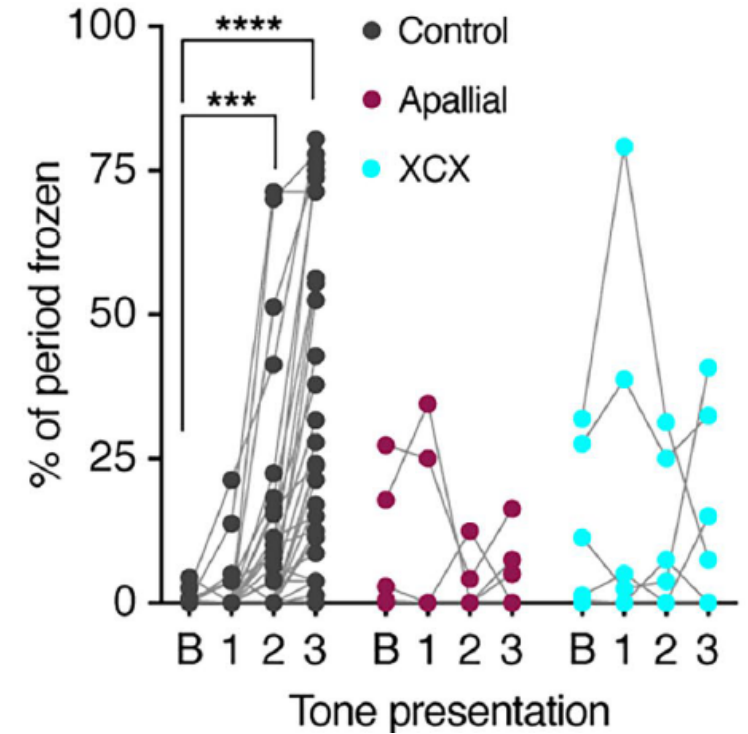
XCX: Xeno (foreigner) + Cortication (cerebral cortex formation)
hCO: human cerebral organoids

Behavioral tests in mouse models

Open Field Test

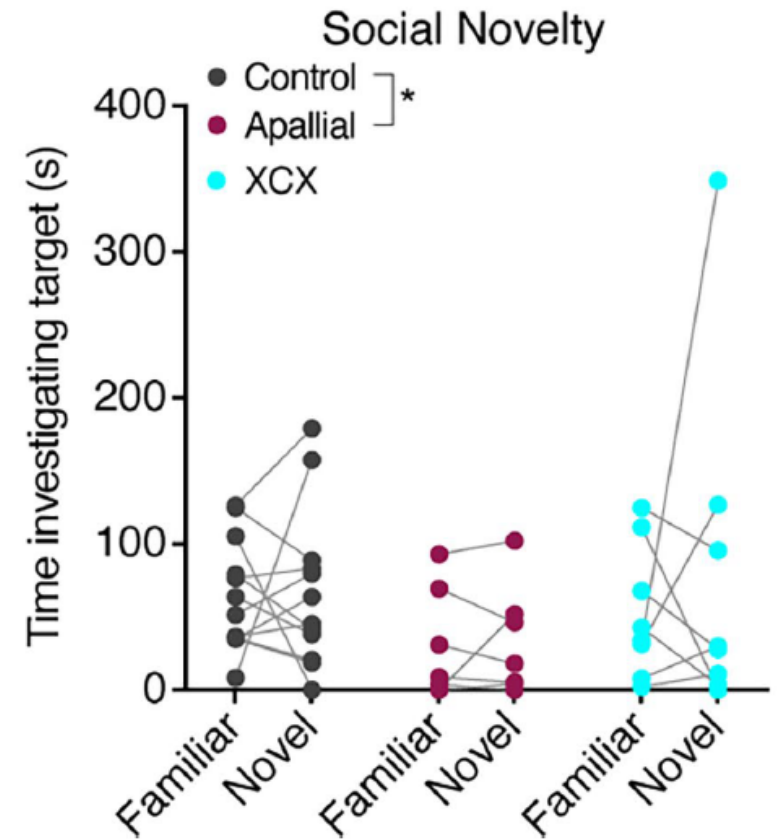
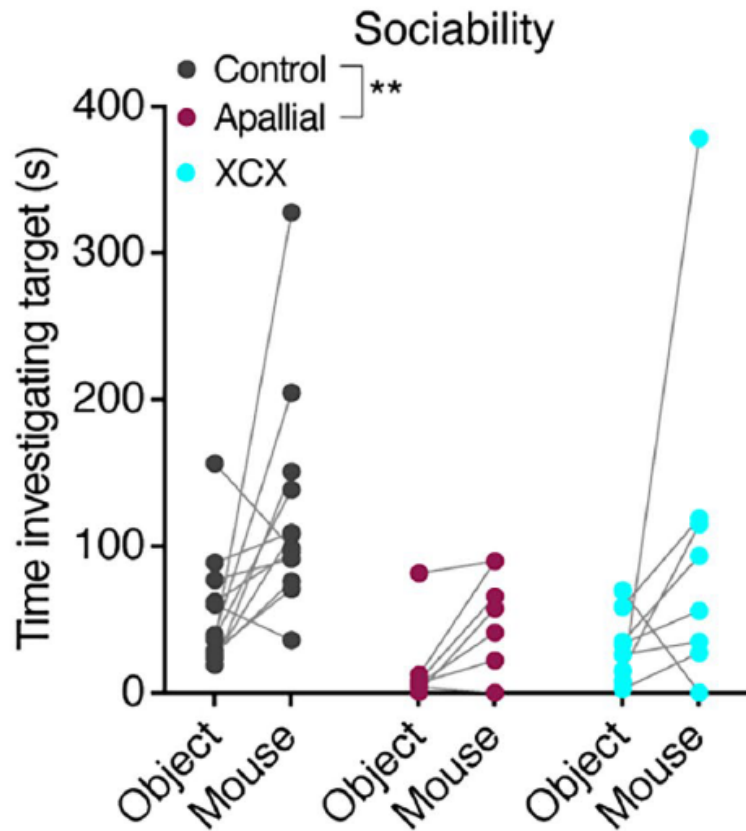


Contextual fear conditioning



Behavioral tests in mouse models

Social tests





Nobel Laureates

- Santiago Ramón y Cajal
- Andrew Huxley and Alan Hodgkin
- Eric R. Kandel
- Roger Y. Tsien
- Emmanuelle Charpentier and Jennifer Doudna
- John B. Gurdon and Shinya Yamanaka

Models used in this lecture

Various animal models

Squid giant axon

Aplysia (Sea slug/ hare)

Aplysia (Sea slug/ hare)

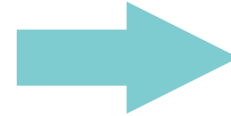
CRISPR/Cas9 (in mouse)

iPSC (in mouse)

Perspectives



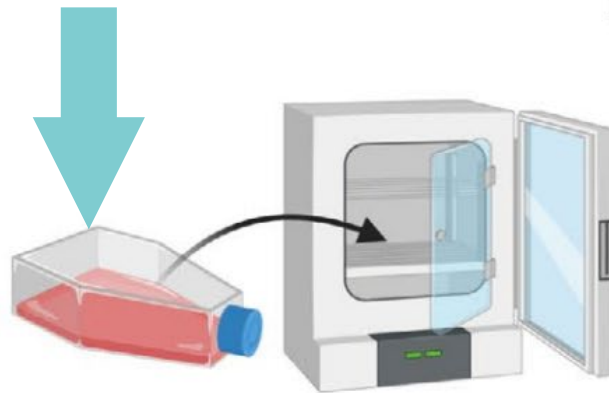
Animal models in neuroscience



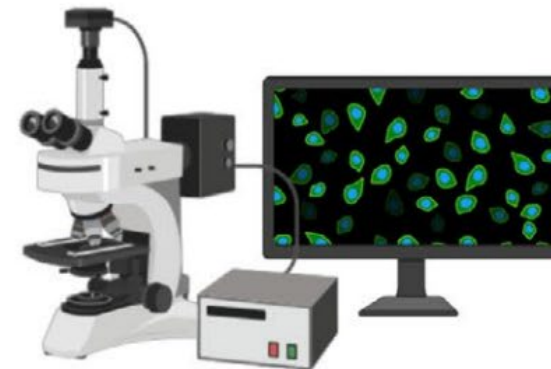
Biomedical perspective



Ethical perspective



In vitro (cell culture, microscopy) techniques



In silico (computational) modeling
Neziri et al., 2024. Animal Model Exp Med.